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September 3, 2019

1 Data

The calibration is performed with a Ra-226 alpha source. Files with the raw experimental data used for the calibration:

- Measurement with the 1 mm sensor from the left telescope only dsd_l_02_0001...
- Measurement with the 20 μm sensor in front of the 1 mm sensor from the left telescope dsd_20_l_03...
- Measurement with the 1 mm sensor from the right telescope only dsd_r_01_0001...
- Measurement with the 20 μm sensor in front of the 1 mm sensor from the right telescope dsd_20_r_02...

2 The noise suppression thresholds

The algorithm used for cutting off the low-amplitude noises is implemented in the macro and illustrated below by the example of the 8th Y-strip (from the measurement dsd_l_02_0001..). The original spectrum is shown in the figures 1-2

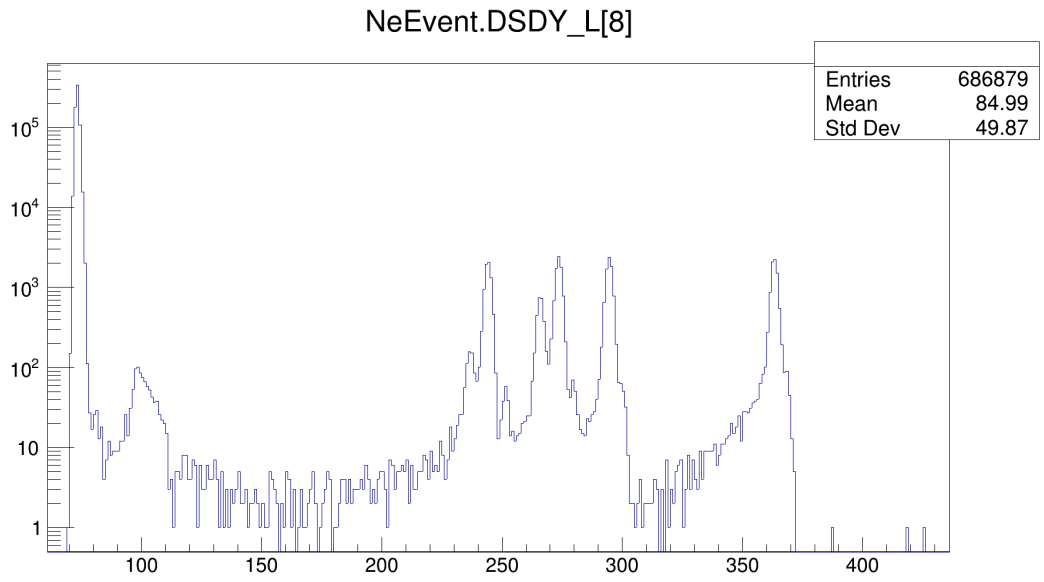


Figure 1: The raw spectrum of the 8th strip of the 1-mm sensor from the left telescope. Measurement without a thin sensor

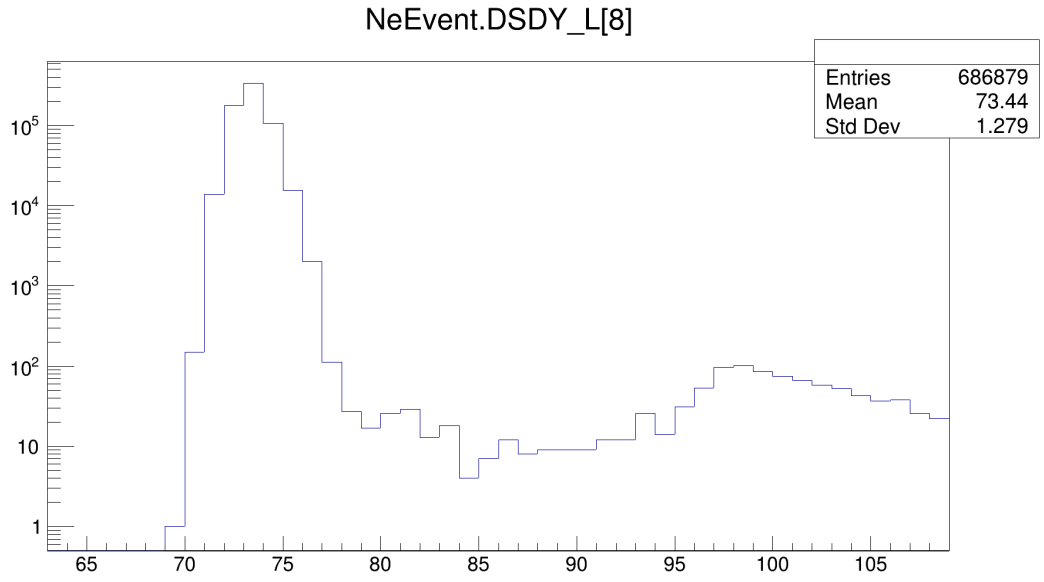


Figure 2: Zoom of the low-amplitude part of the spectrum shown in the figure 1

The threshold finding algorithm consisted of two steps:

1. To find a bin with the highest count (74th in the figure 2);
2. To increase the bin number until the bin height stops decreasing. (79th in the figure 2);

In the same way, the thresholds for all the readout channels are found. Then event selection by the hit multiplicity equal to one is performed.

3 Assessment of the quality of the data

In order to characterize the quality of the calibration data the ratio of the RMS of the high energy peak (calculated for an 8 bin window) to the difference between the ADC values of the highest and the lowest energy peaks was used as a figure of merit (FOM).

The FOMs for the X and Y strips of the 1-mm sensor of the left telescope are shown in the figures 3 and 4 respectively.

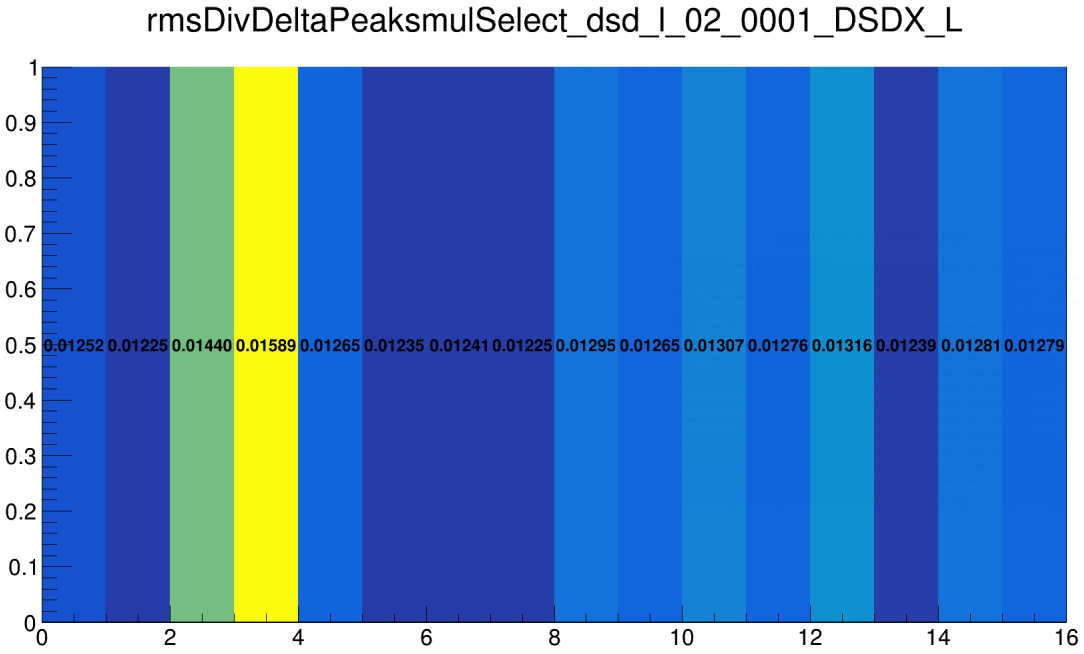


Figure 3: Ratio of the RMS value in a high energy peak to diffrence of ADC-channels in the first and the last peaks from the X-strips of the left telescope

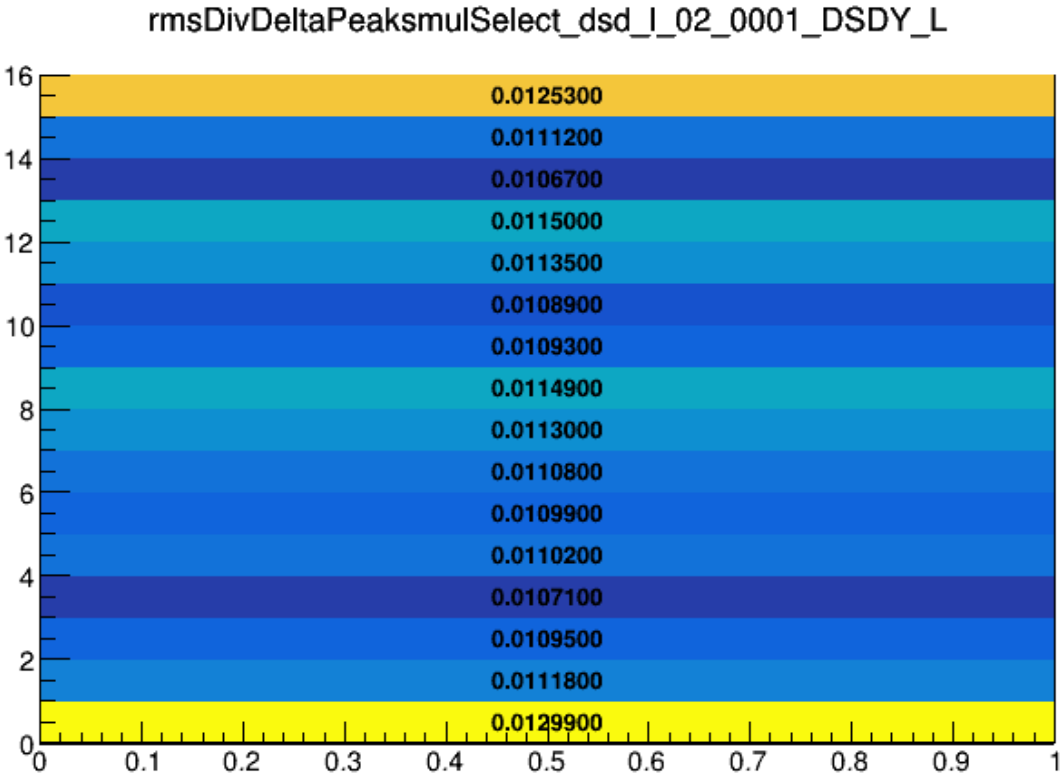


Figure 4: Ratio of the RMS value in a high energy peak to diffrence of ADC-channels in the first and the last peaks from the Y-strips of the left telescope

One can see that for the Y-side strips the FOM is smaller than for the X-side strips.

We have selected the Y-side for the detailed investigation.

4 Evaluation of the front dead layer

In order to determine the dead layer a special dimensionless parameter η is introduced $\eta(d) = \dfrac{E_3 - E_2}{E_3 - E_1}$, where E_1, E_2 and E_3 are residual kinetic energies of the alpha-particles after propagation through the layer of silicon with the thickness d. Initial energies of alfa-particles are 4.7844 MeV, 6.0024 MeV and 7.6869 MeV respectevaly.

A GEATN4 Monte-Carlo simulation is used for obtaining the $\eta(d)$ dependence. The simulations are performed for the following values of the dead

layer thickness (Si- equivalent): 2 μm, 4 μm, 5 μm, 5.56 μm and. The residual energies of the alfa-particles after passage through the dead layer are shown in the table 1.

Table 1: Energy of the α-particle after the propagation throught different dead layer thickness, GEANT4 simulation

Thickness, [μm]	E_1, [MeV]	E_2, [MeV]	E_3, [MeV]
0	4.7844	6.0024	7.6869
2	4.4815	5.7408	7.4651
4	4.1671	5.4738	7.2416
5	4.0007	5.3348	7.1274
5.56	3.9109	5.2573	7.0651

The linear fit of the η parameter $\eta(d) = md + n$ is shown in the figure 5.

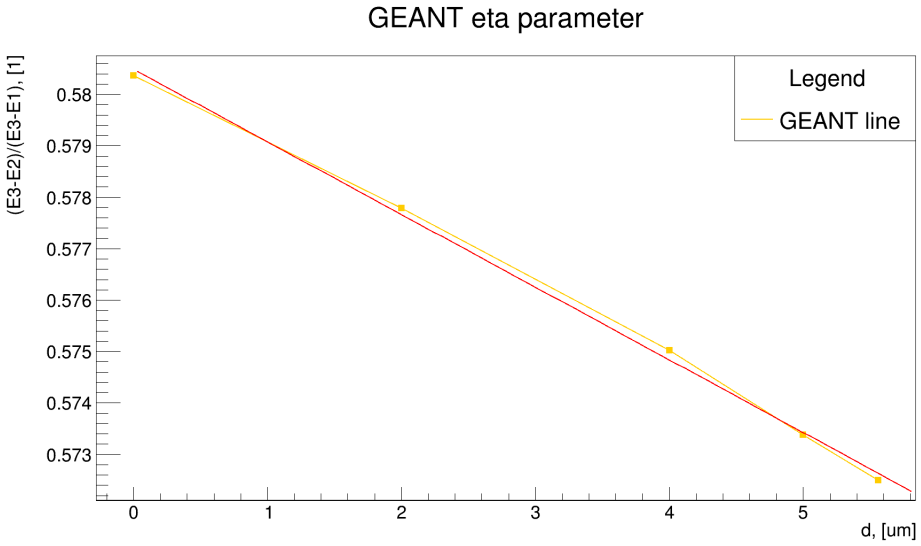


Figure 5: The fit of the $\eta(d)$ parameter (red line)

Table 2: Energy of the α-particle after the propagation throught different dead layer thickness, LISE++ simulation

Thickness, [μm]	E_1, [MeV]	E_2, [MeV]	E_3, [MeV]
0	4.7844	6.0024	7.6869
2	4.4798	5.7424	7.4643
4	4.1602	5.4719	7.2365
5	3.9948	5.3327	7.1207
5.56	3.9012	5.2536	7.0553

The residual energies were also found using the LISE++ programme. Comparing the results from GEANT4 and LISE++ one can get an idea about the accuracy of the η parameter determination. The residual energies of the alfa-particles after passage through the dead layer are shown in the table 2. The $\eta(d)$ dependencies obtained with GEANT4 and LISE++ are drawn in the figure 6.

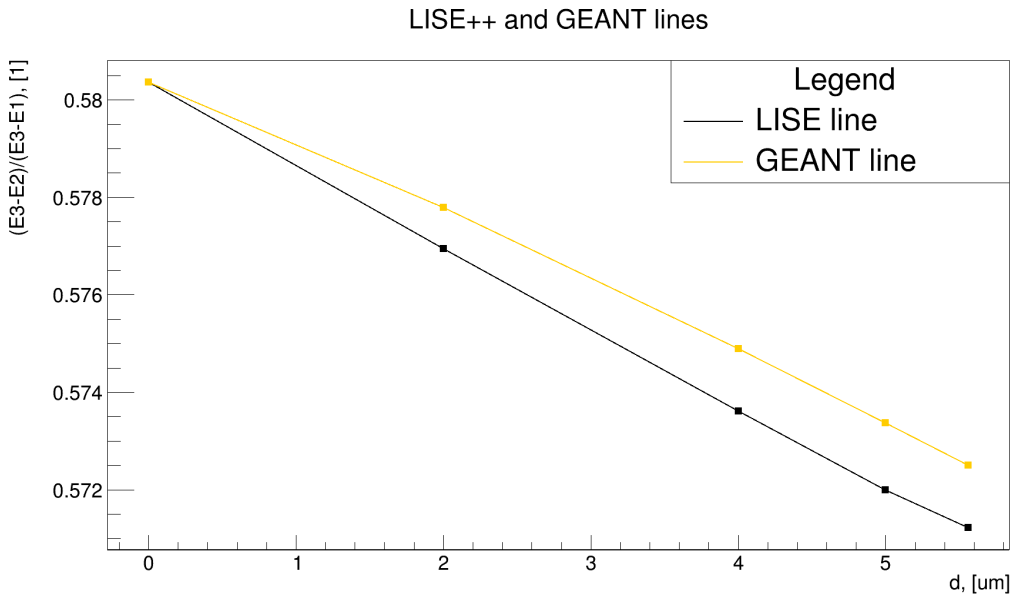


Figure 6: $\eta(d)$ dependencies obtained with GEANT4 and LISE++

We assume that calibration is linear $E(N) = aN + b$, so η parameter for the experimental spectrum is $\eta_{exp}(d) = \dfrac{N_3 - N_2}{N_3 - N_1}$, where N_i is a corresponding ADC peak.

Using coefficients from the linear fit (figure 5) we can evaluate dead layer thickness as $d = \dfrac{\eta(d) - n}{m}$.

The results of the front dead layer evaluation based on the data from the Y-side and X-side of the 1-mm-thick sensor from the left telescope are shown in the figures 7 and 8 respectively.

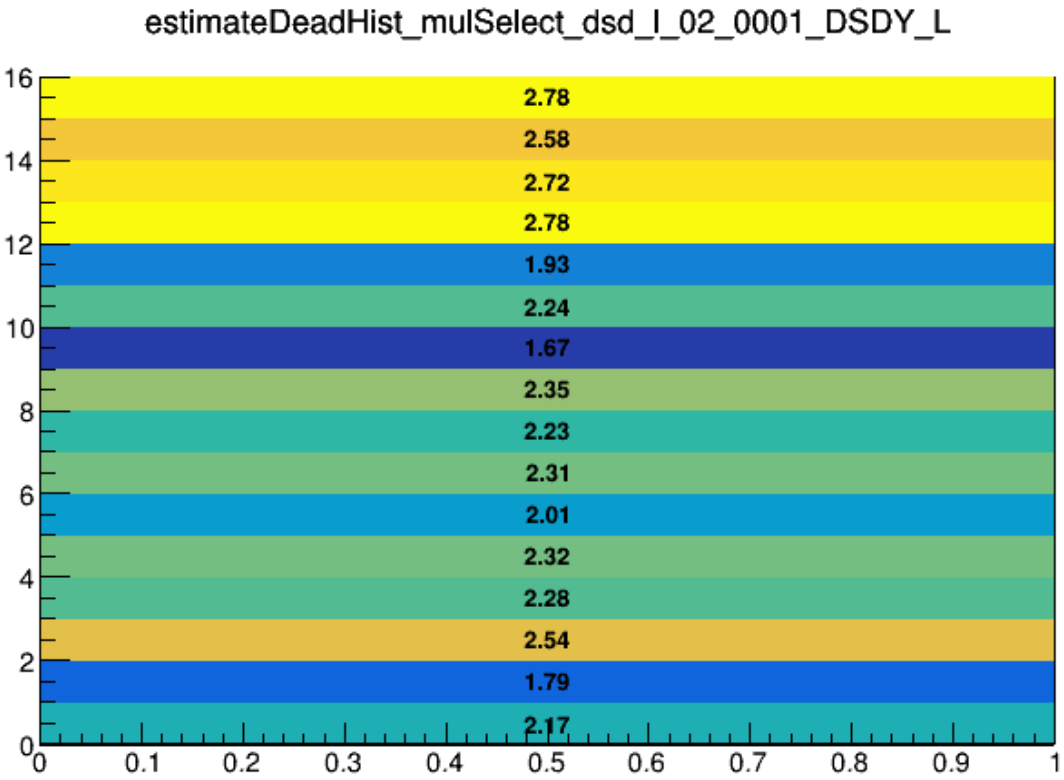


Figure 7: Dead layer thickness estimation by the Y-strips. The mean value is 2.29 μm. The left telescope

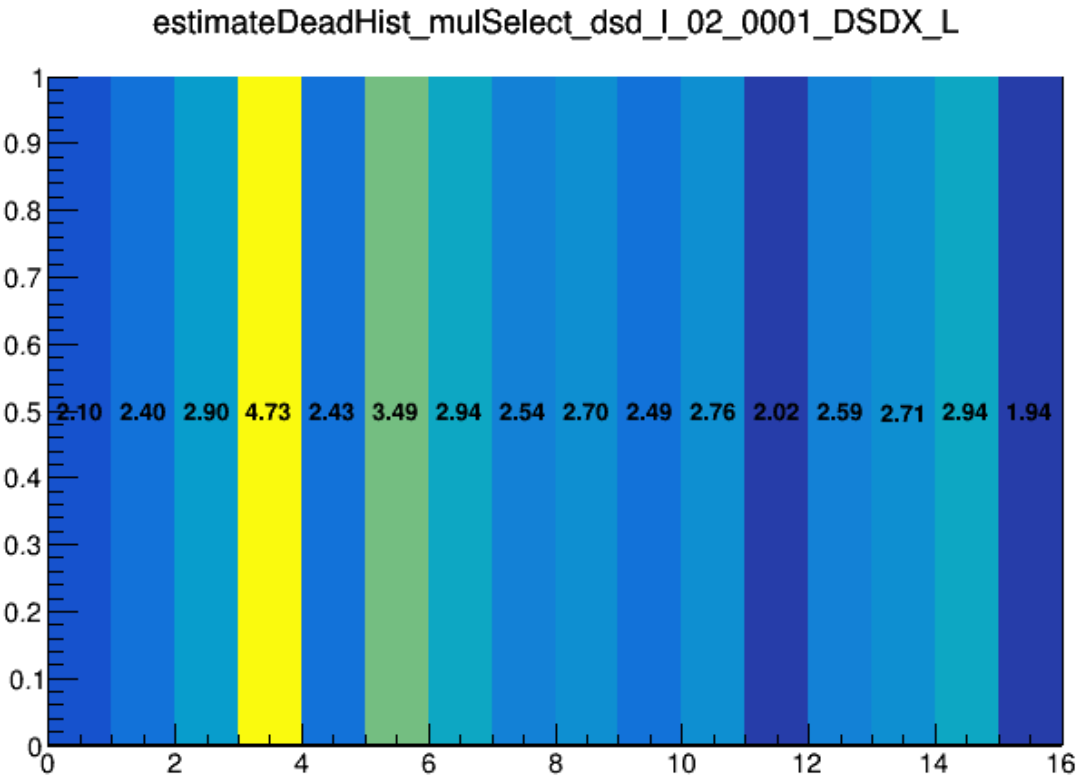


Figure 8: Dead layer thickness estimation by the X-strips. The mean value is 2.47 μm. The left telescope

5 Calibration of the left telescope

The calibration can be done after taking into account the energy losses of the alpha-particles in the dead layer.

The Monte Carlo simulated dependence of the energy loss on the dead layer thickness (Si equivalent) was fitted by a quadratic function $dE(d) = p_2 d^2 + p_1 d + p_0$, where d is the thickness of the dead layer [μm], p_i are the fitting parameters. An example of such fitting for the energy 7.69 MeV is shown in the figure 9. The results of the fitting are shown in the table 3.

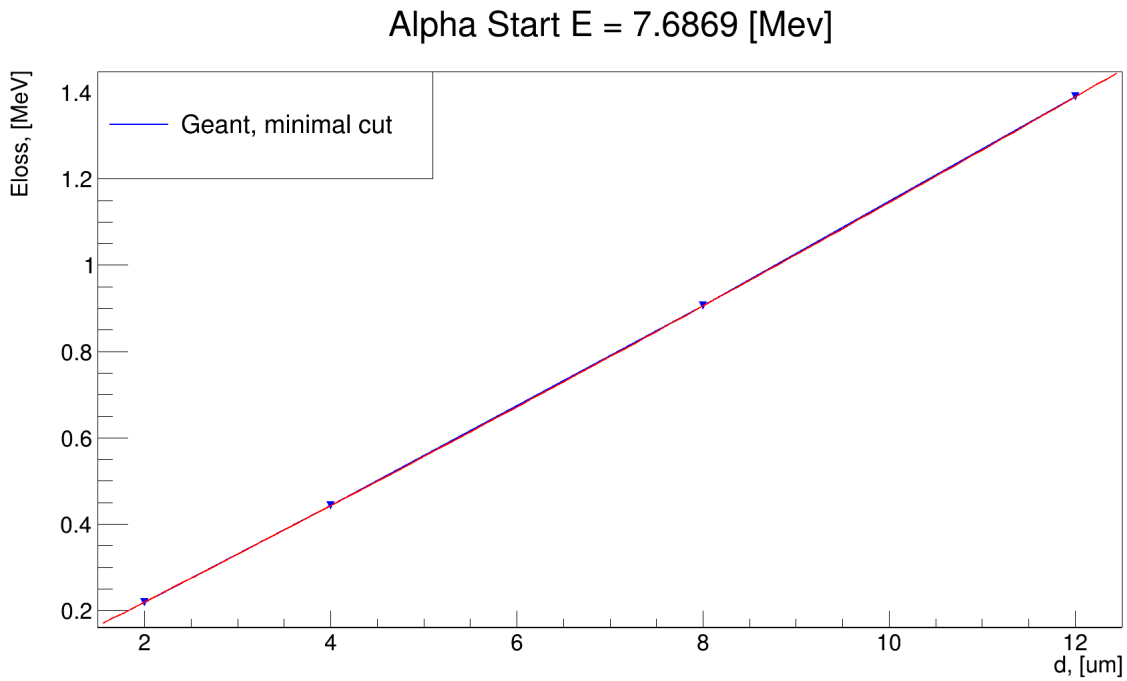


Figure 9: Quadratic fit (red line) of the simulated graph $E(d)$ (blue line)

Table 3: Parameters of the quadratic fit for energy losses in the dead layer.

Start energy,[MeV]	p_2	p_1	p_0
4.7844	0.0113729	0.141487	0.00241099
6.0024	0.00436517	0.125545	0.00131201
7.6869	0.00178522	0.107627	0.00067292

We assumed that the dead layer is uniform. The average dead layer thickness value from the figure 7 was used for calculating the energy released in the active volume of the sensor by alpha particles of different initial energies

The peak position finding algorithm is discussed in the appendix.

The linear fit of the $E(N) = aN + b$, dependence was used for determination of the calibration parameters (figure 10). Here E is the average energy of an alpha particle after propagation through the dead layer, N is the position of corresponding ADC peak, a and b — are the calibration coefficients.

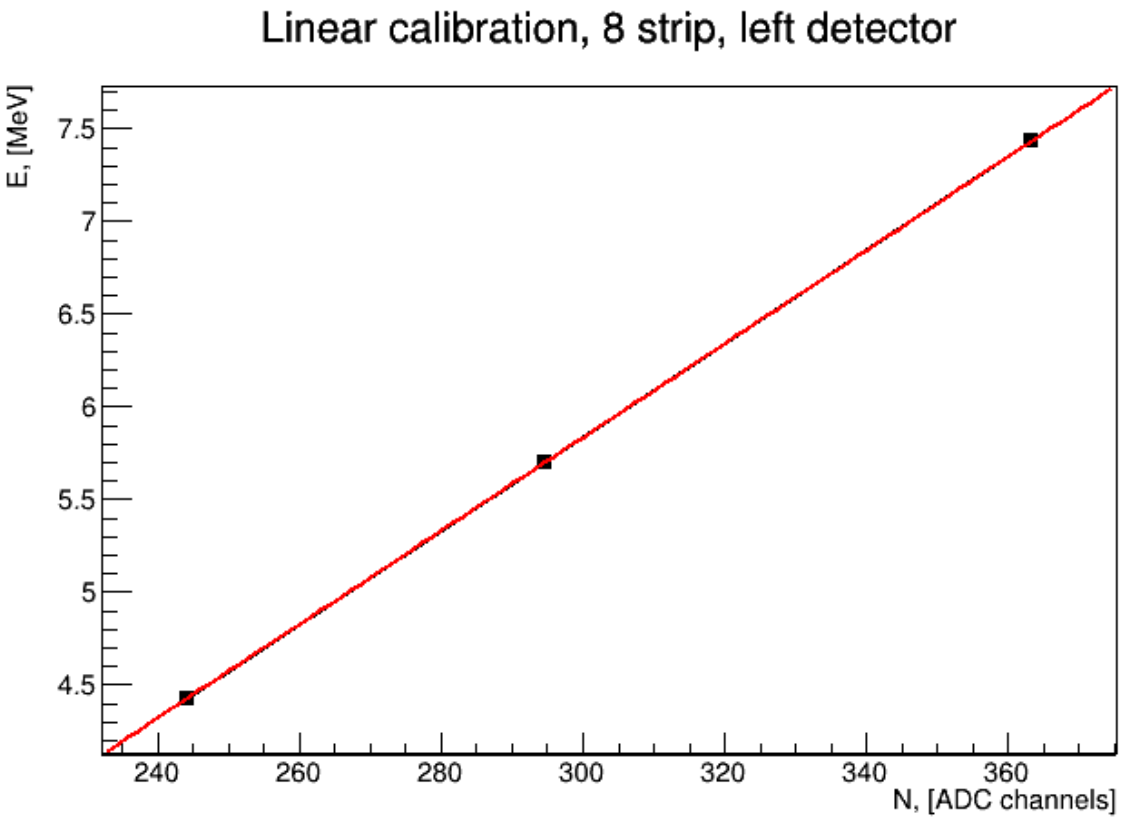


Figure 10: Calibration plot $E(N)$ for the 8th strip of the 1-mm-thick sensor. The left telescope

Calibration coefficients for the Y side of the 1-mm-thick sensor of the left telescope are listed in the table 6.

Table 4: Calibration coefficients for the Y side of the 1-mm-thick sensor of the left telescope.

Strip number	a	b
0	0.0248743	-1.83827
1	0.0251701	-2.37063
2	0.0246235	-1.86079
3	0.0247815	-2.2852
4	0.0246906	-1.98389
5	0.0250748	-2.34883
6	0.0249042	-1.97226
7	0.0251791	-2.27259
8	0.0252418	-1.73421
9	0.0245996	-2.36944
10	0.0246349	-1.82582
11	0.0245234	-2.13858
12	0.0252339	-1.71312
13	0.0247386	-2.26129
14	0.0255358	-1.81593
15	0.0245782	-2.15246

6 Building the thickness map of the 20-μm-thick sensor of the left telescope

The thickness of the 20-μm-thick sensor was measured in each pseudo-pixel, i.e. an intersection of a strip from the 20-μm-thick sensor and a perpendicular strip from the 1-mm-thick sensor of the same telescope.

The thickness was derived from the shift of the peak measured in the 1-mm-thick sensor and corresponding to the highest energy alpha-particles when the 20-μm-thick sensor is mounted in front of the 1-mm-thick sensor.

The sample spectra from the 1-mm-thick sensor with(the dsd_20_l_03.. is used) and without (the dsd_l_02_0001.. is used) the 20-μm-thick sensor are shown in figures 13 and 11 respectively. Zoomed peaks are shown in the figures 14 and 12 respectively. Note that for measurements without the 20-μm-thick sensor the spectrum for the entire strip is shown while for the measurements with the the 20-μm-thick sensor the spectrum is shown for a pseudopixel.

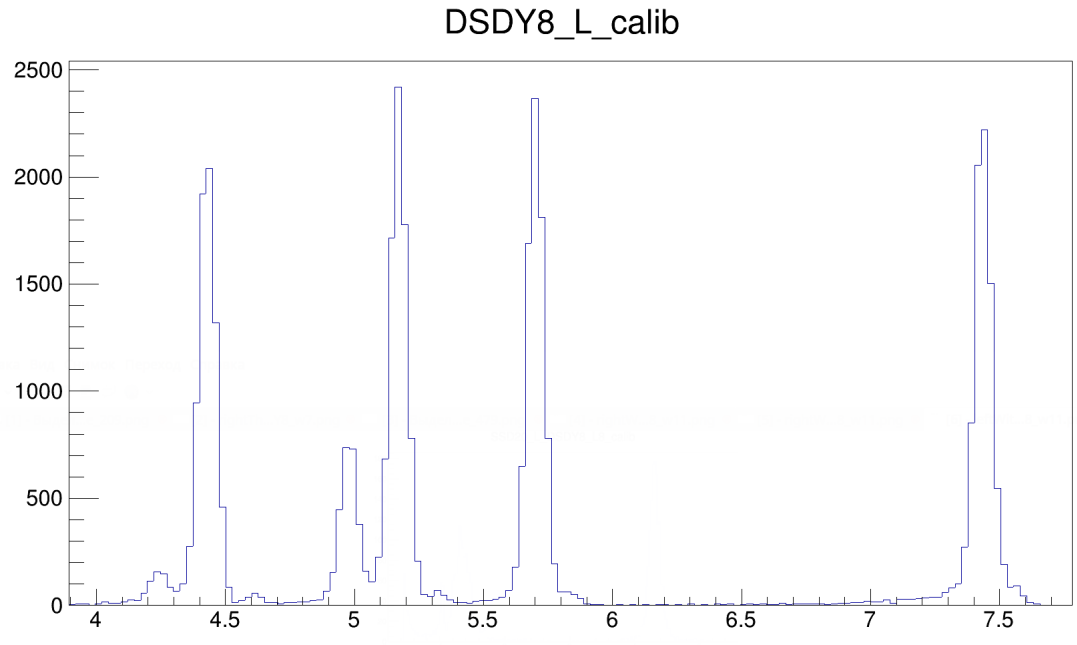


Figure 11: The full spectrum of the 8th Y-strip, the measurement without the left 20-μm-thick sensor

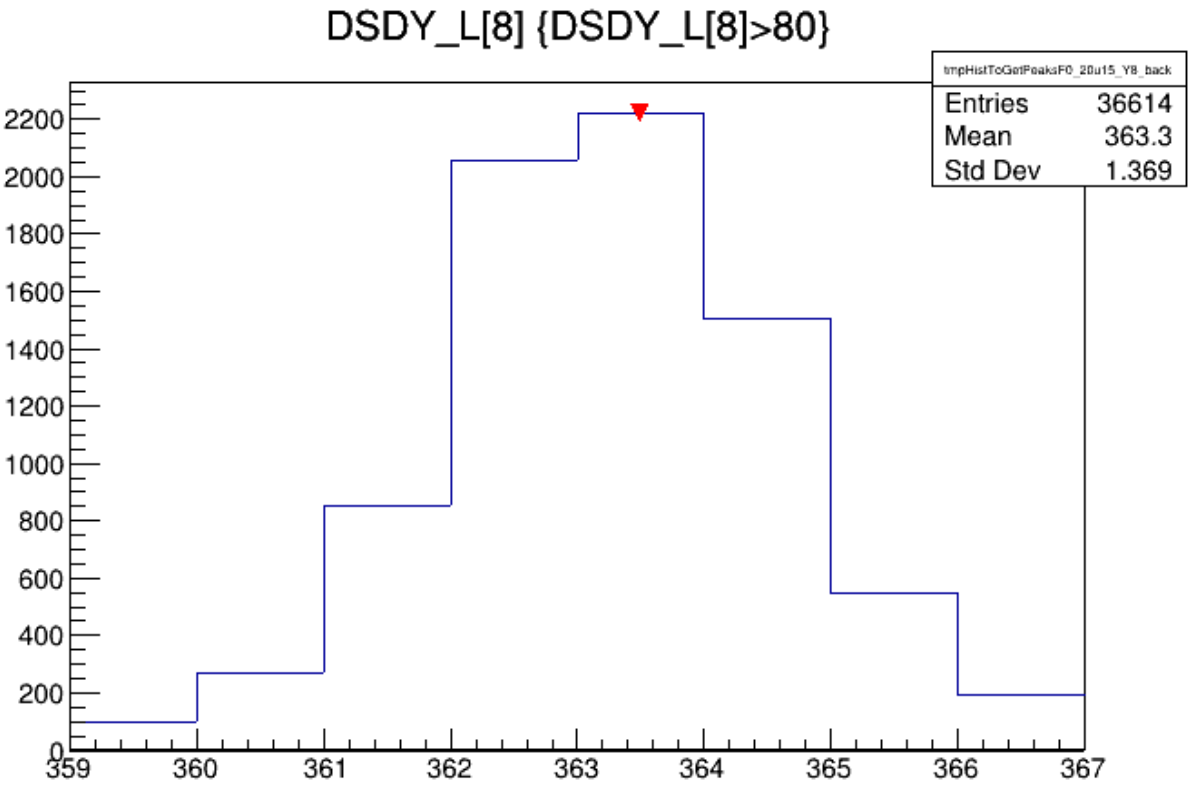


Figure 12: An example of the highest energy peak for the 8th Y-strip, the measurement without the left 20-μm-thick sensor

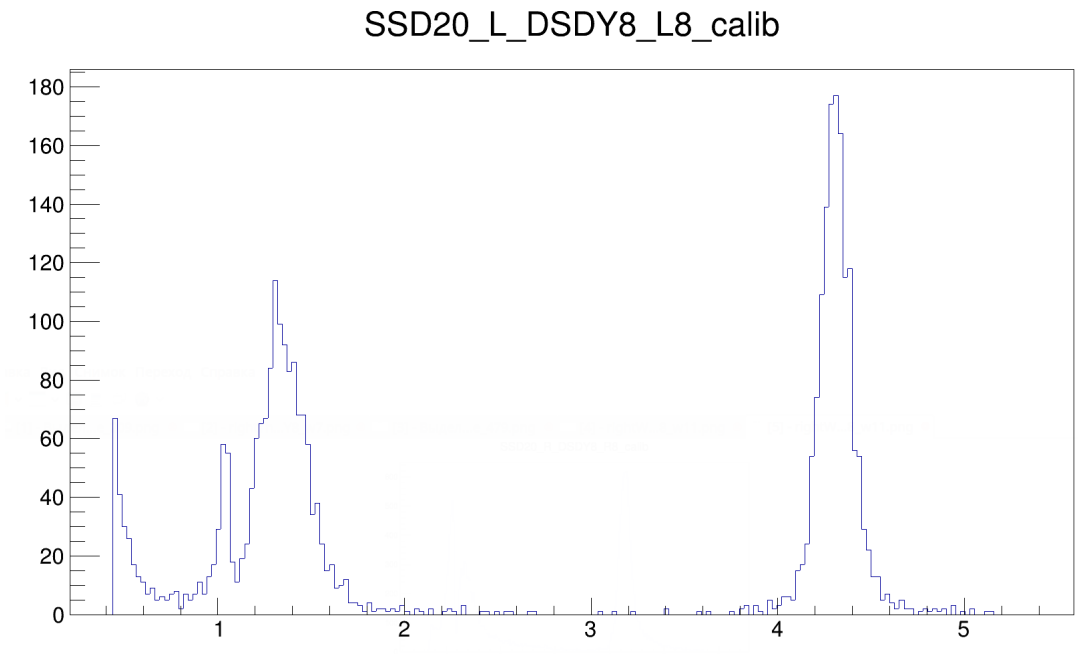


Figure 13: The full spectrum of the pseudopixel 20u8-Y8 in the measurement with left 20-μm-thick sensor

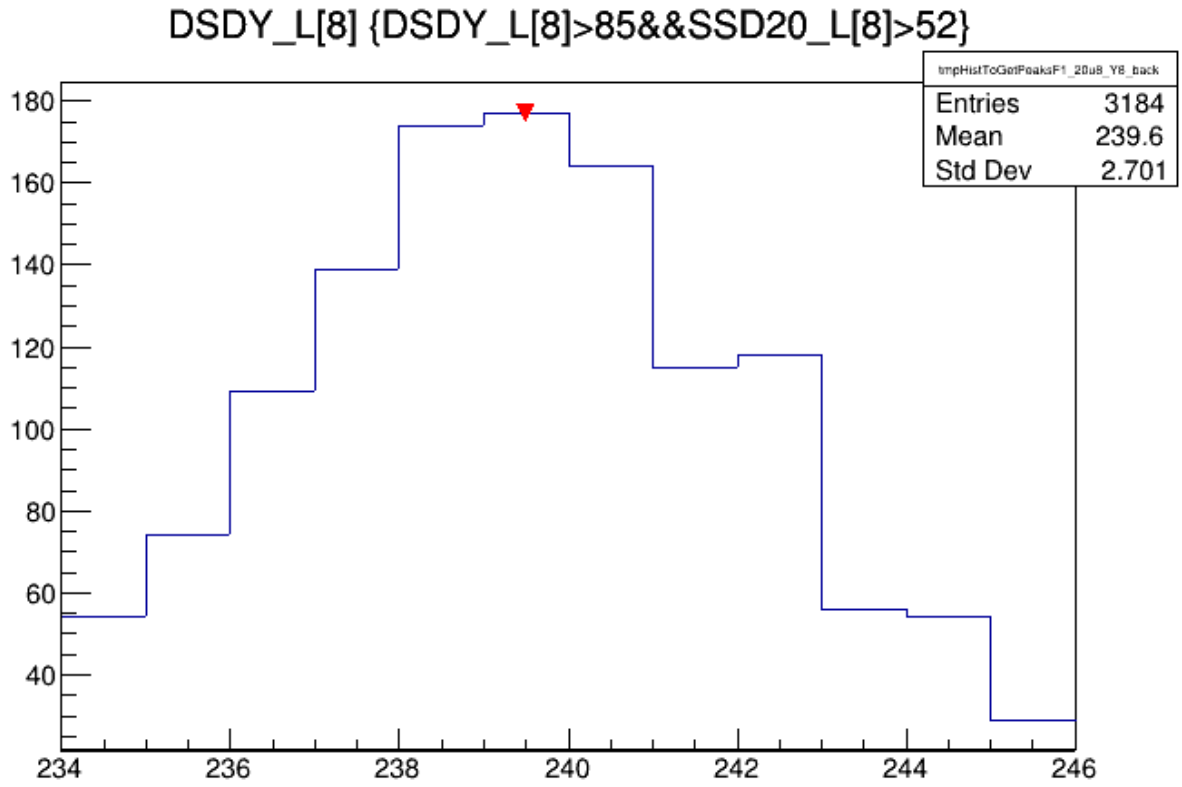


Figure 14: An example of the highest energy peak in the pseudopixel 20u8-Y8, in the measurement with the left 20-μm-thick sensor

The maps of the highest energy peak values are shown in the figures 15 - 16.

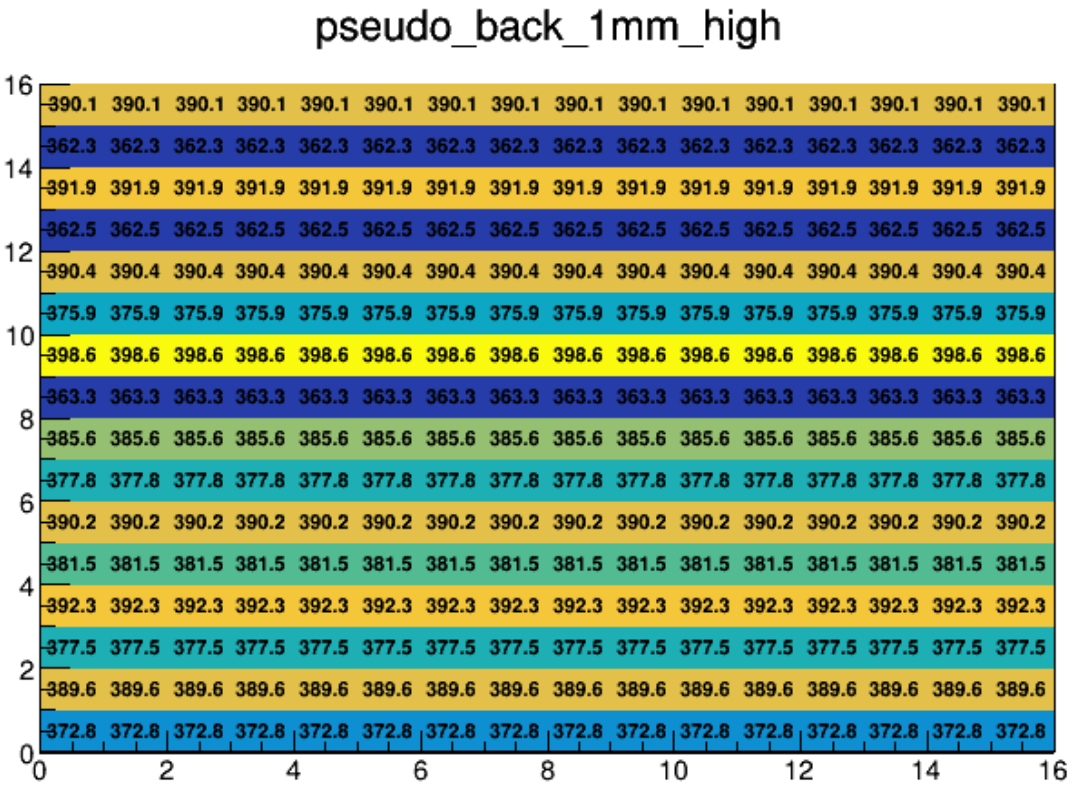


Figure 15: Map of the highest energy peak position in the left 1-mm-thick sensor without the left 20-μm-thick sensor

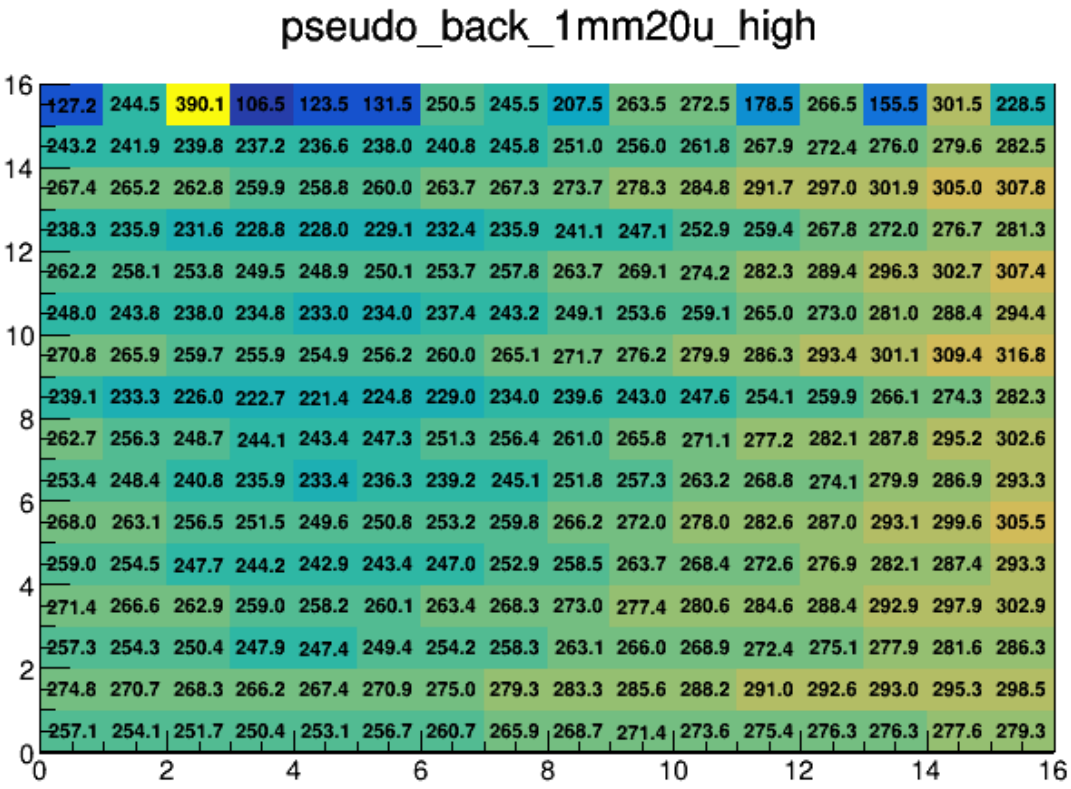


Figure 16: Map of the highest energy peak position in the left 1-mm-thick sensor with the 20-μm-thick sensor

For each pseudopixel, the shift of the highest energy peak was converted into the thickness using parameters like in the table 3 but for the fitting data extended until the 32 μm thickness. Thus, $E(d)$ - relation of the alpha-particle energy after the propagation throught a silicon layer looks like in the figure 17 and fit parameters are listed in table 5.

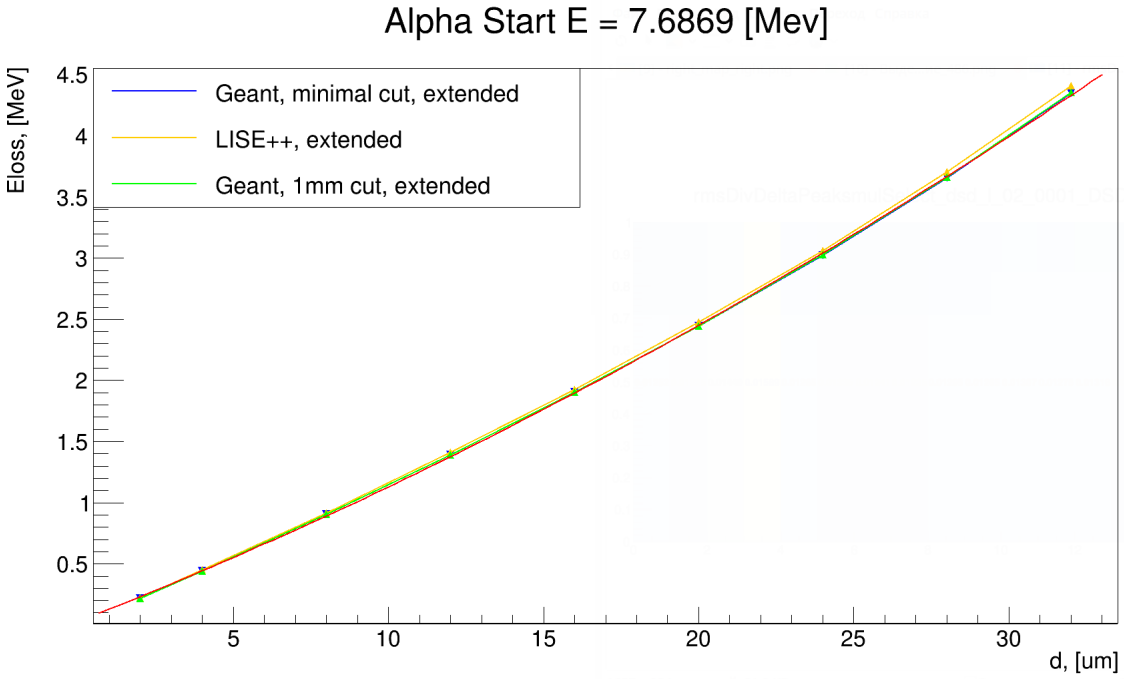


Figure 17: Quadratic fit (red line) of the extended simulated data $E(d)$ for the high enegry

Table 5: Parameters of the quadratic fit for energy losses estimations in the dead layer.

Start energy,[MeV]	p_2	p_1	p_0
7.6869	0.00111028	0.0988807	0.0297804

The algorithm of the pixel thickness search in the thin sensor:

1 An energy in the thick sensor is calculated

$$E = aN + b,$$

where N — ADC channels value of the high energy peak, a and b — calibration parameters. This is a peak position after propagation through the thin sensor and dead layer of the thick one.

2 Energy losses in the thin sensor and dead layer of the thick one

$$dE = E_0 - E,$$

where is E_0 — start energy of the alpha-particle (7.6869 MeV).

3 According to the given approximation full passive thickness d is evaluated.

4 The previously determined thickness of the thick sensor dead layer is subtracted from the full passive thickness to find the pixel thickness of d_{thin} of the thin sensor

$$d_{thin} = d - d_{thickDead},$$

where $d_{thinDead}$ — dead layer of the thick sensor.

The result is shown in the figure 18.

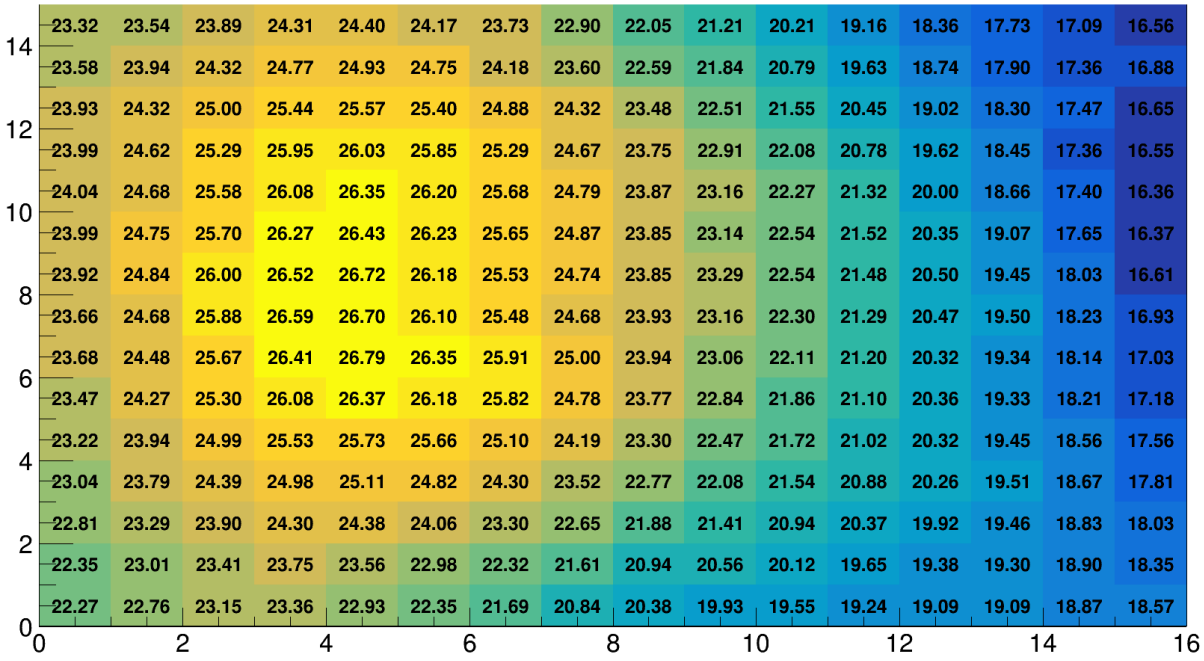


Figure 18: Pseudo-pixel thickness map for the 20-μm-thick sensor of the left telescope

Evaluation of the dead layer on the intersection with the 15th Y-strip isn't shown because we see uneven data in the figure 16. The reason is the frame of the thin sensor that covers part of the thick sensor area.

7 The calibration of the right-side telescope

The following data files are used for the right-side telescope: dsd_r_01_0001.. and dsd_20_r_02.. The pictures and tables below are similar to those shown

above for the left-side telescope.

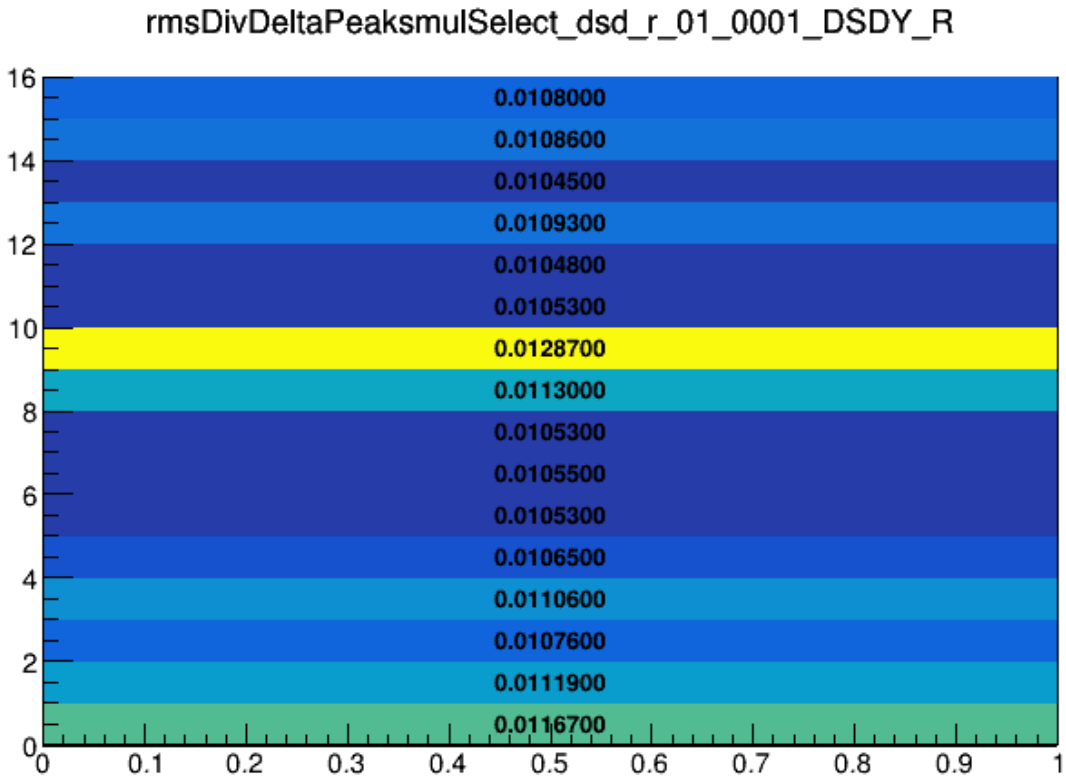


Figure 19: Ratio of the RMS value in a high energy peak to diffrence of ADC-channels in the first and the last peaks from the Y-strips of the right telescope

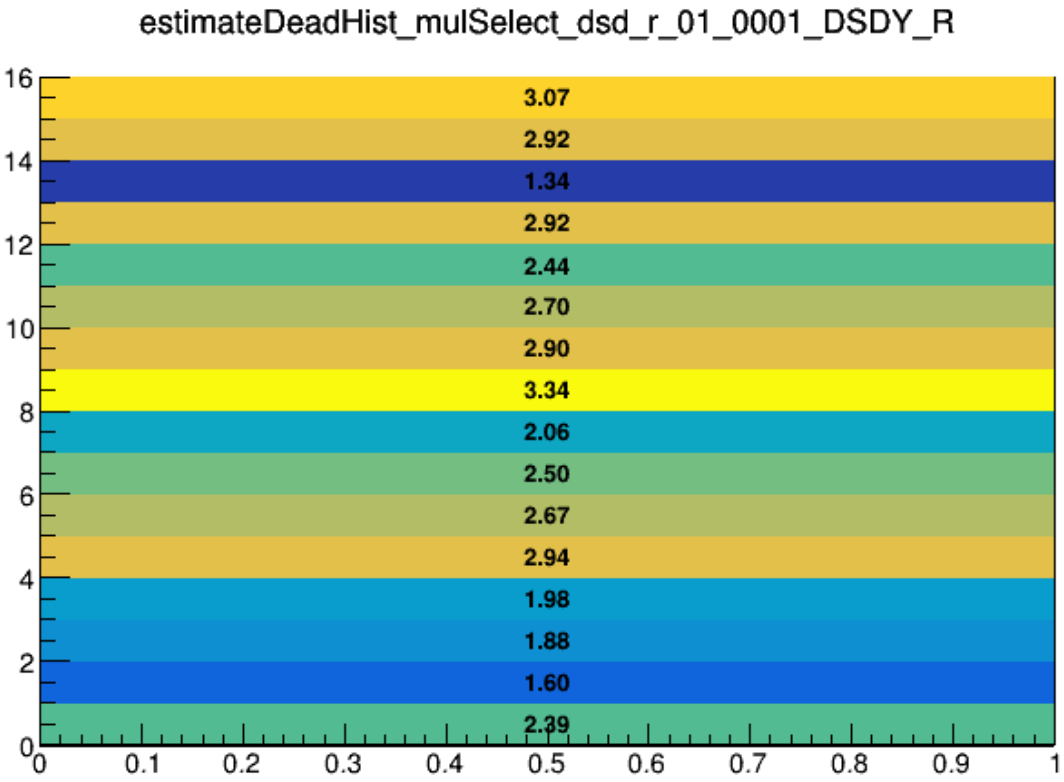


Figure 20: Dead layer thickness estimation by the Y-strips of the right telescope. The mean value is 2.48 μm

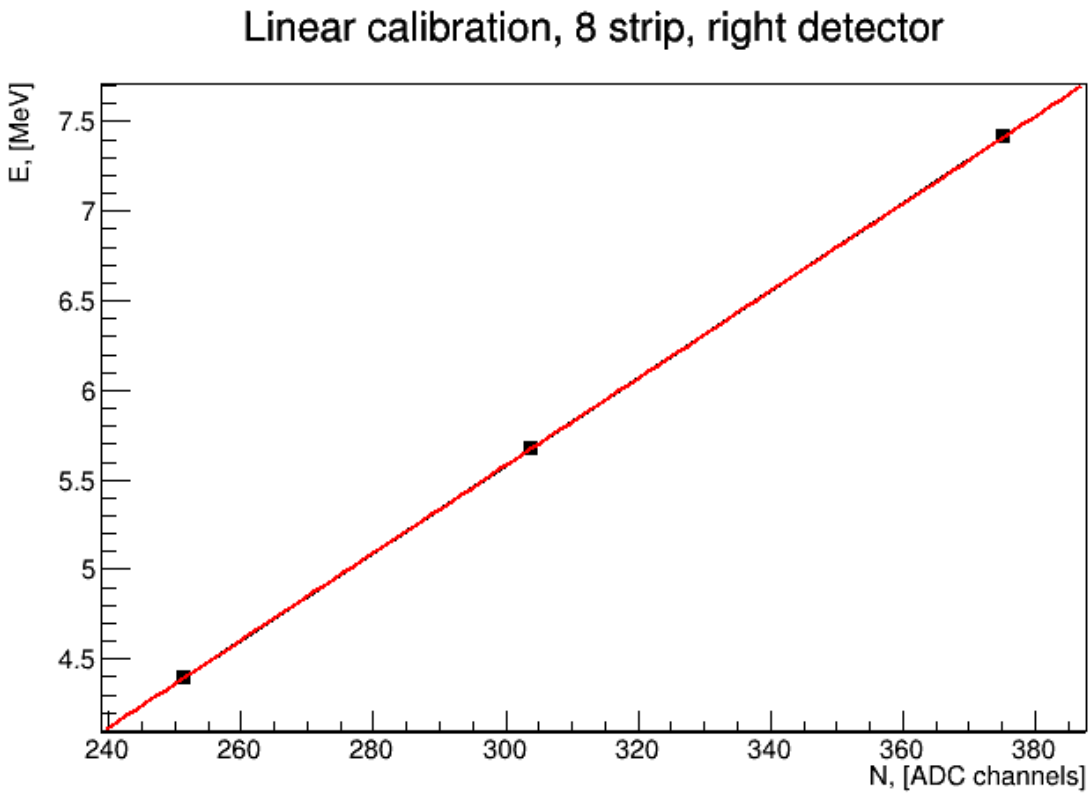


Figure 21: Calibration plot $E(N)$ for the 8th strip of the 1-mm-thick sensor. The right telescope

Table 6: Calibration coefficients for the 1-mm-thick sensor of the right telescope, derived from the Y-strip data.

Strip number	a	b
0	0.0241594	-1.66383
1	0.0248167	-1.66527
2	0.0244318	-1.59953
3	0.0242835	-1.70919
4	0.0243111	-1.6488
5	0.0242424	-1.69636
6	0.0240563	-1.74666
7	0.0236885	-1.63158
8	0.0243536	-1.72497
9	0.0245008	-1.98168
10	0.0239052	-1.62719
11	0.0245641	-1.82977
12	0.0246955	-1.72355
13	0.0246122	-1.9134
14	0.0245884	-1.6579
15	0.0247488	-1.86683

7.1 Building the thickness map of the 20-μm-thick sensor of the right telescope

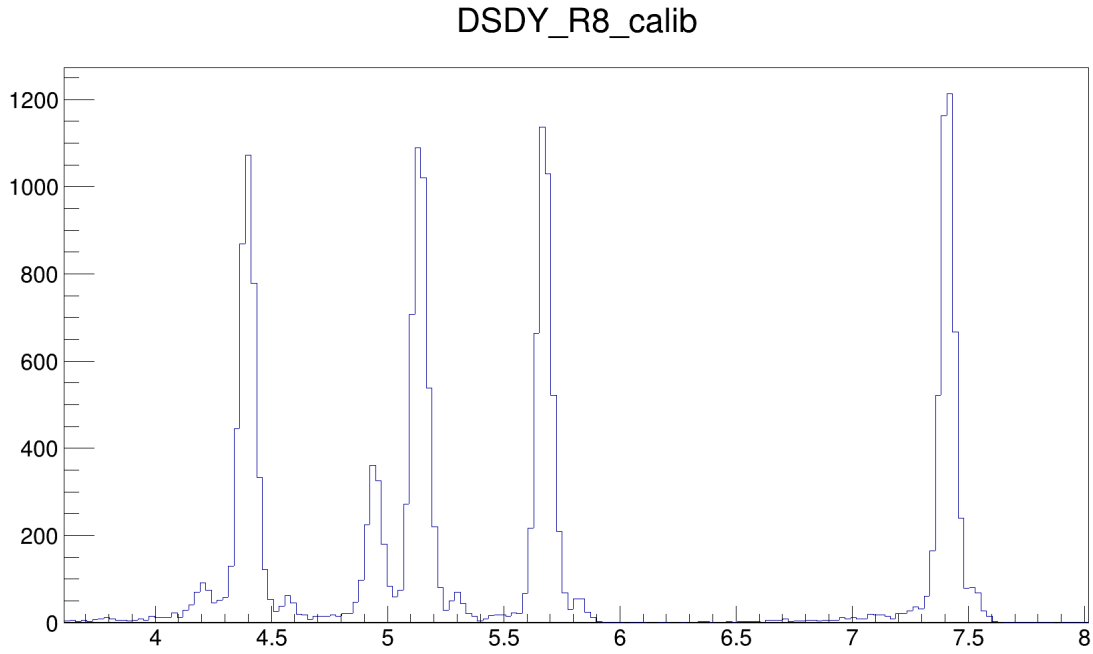


Figure 22: The full spectrum of the 8th Y-strip, the measurement without the right 20-μm-thick sensor

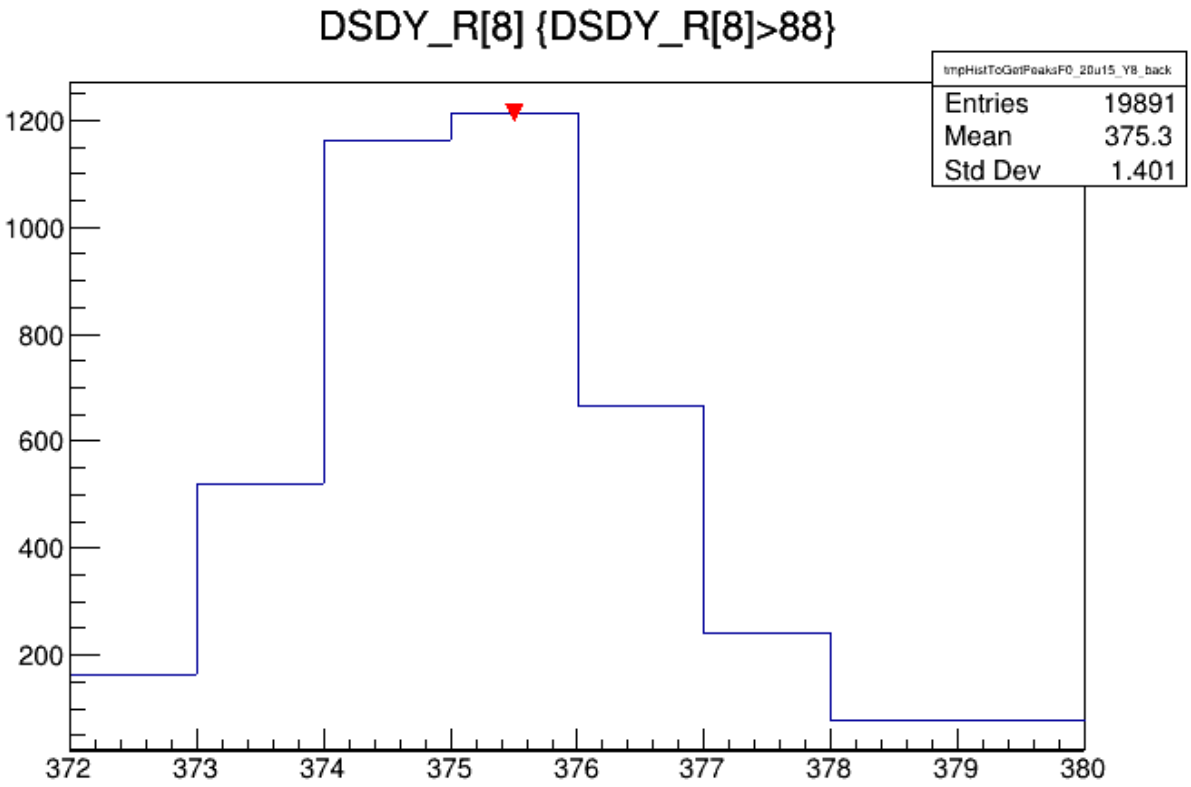


Figure 23: An example of the highest energy peak for the 8th Y-strip, the measurement without the right 20-μm-thick sensor

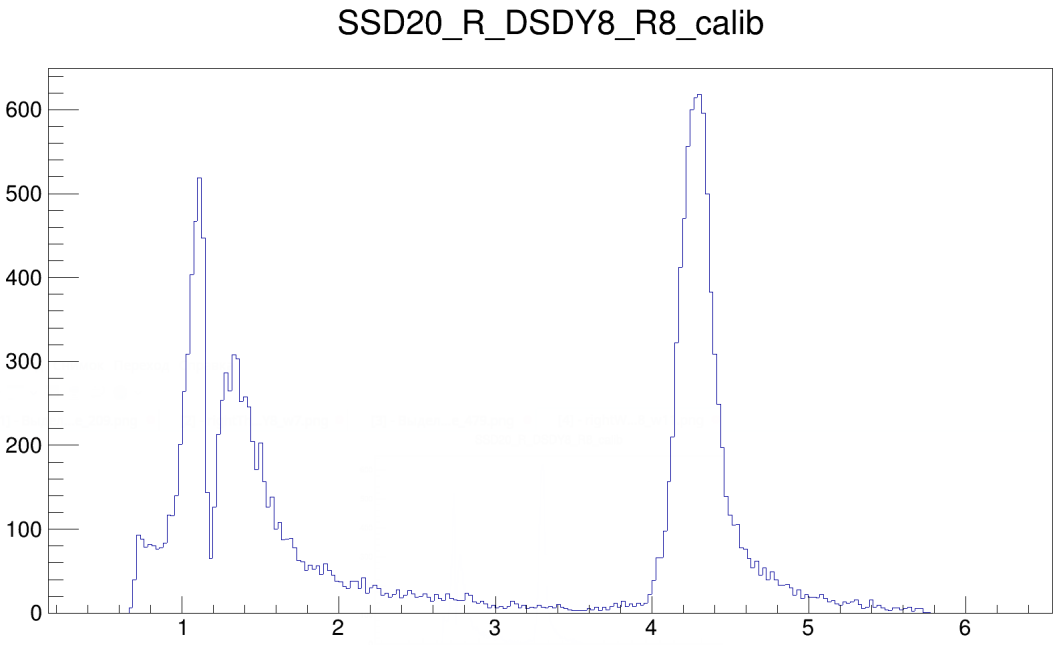


Figure 24: The full spectrum of the pseudopixel 20u8-Y8 in the measurement with the right 20-μm-thick sensor

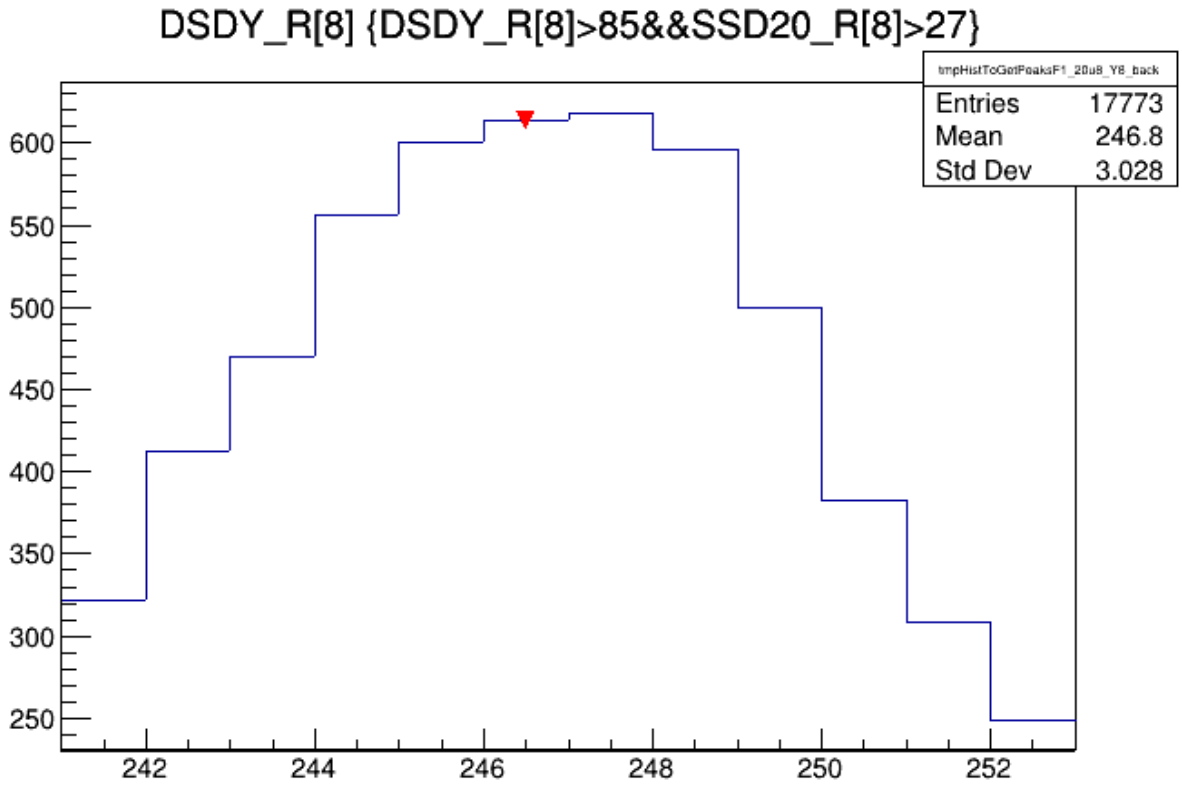


Figure 25: An example of the highest energy peak in the pseudopixel 20u8-Y8, in the measurement with the right 20-μm-thick sensor

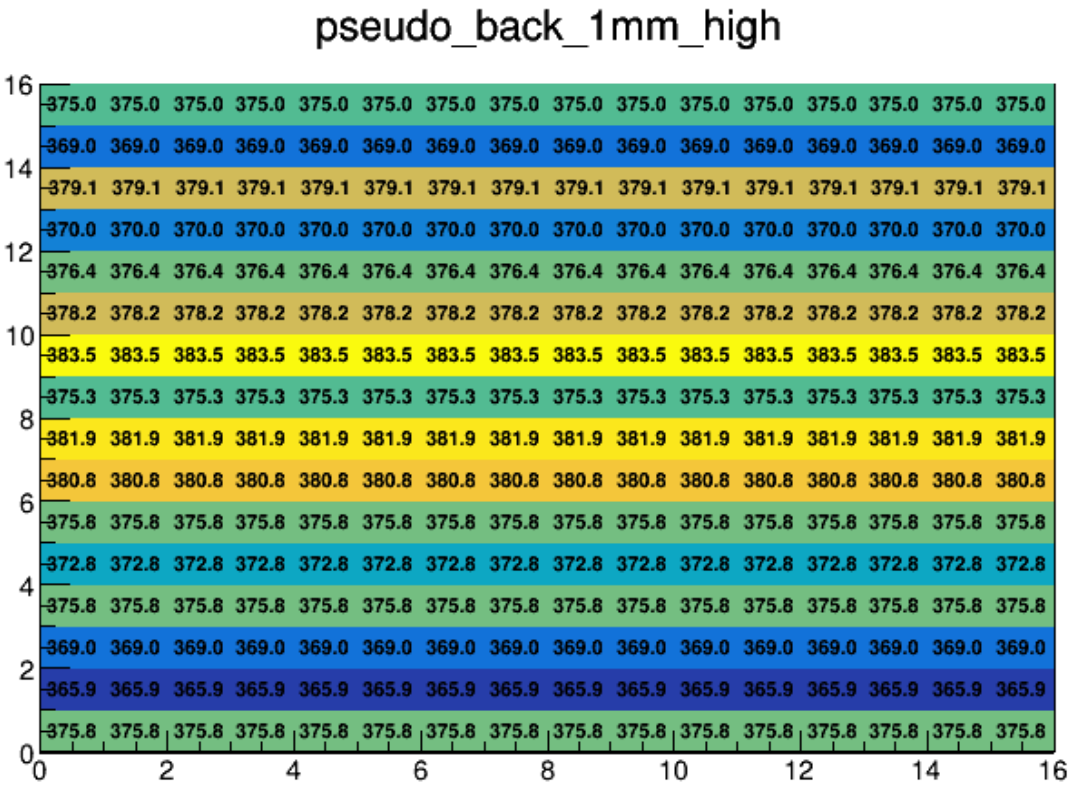


Figure 26: Map of the highest energy peak position in the right 1-mm-thick sensor without the 20-μm-thick sensor

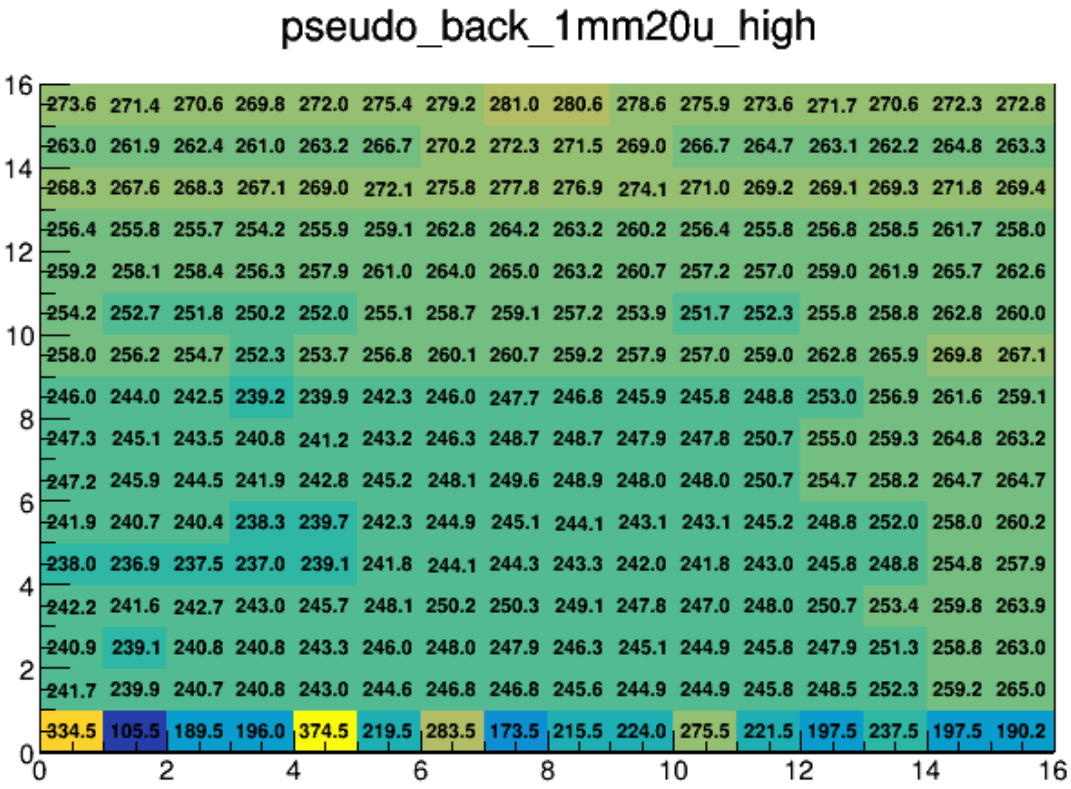


Figure 27: Map of the highest energy peak position in the right 1-mm-thick sensor with the 20-μm-thick sensor

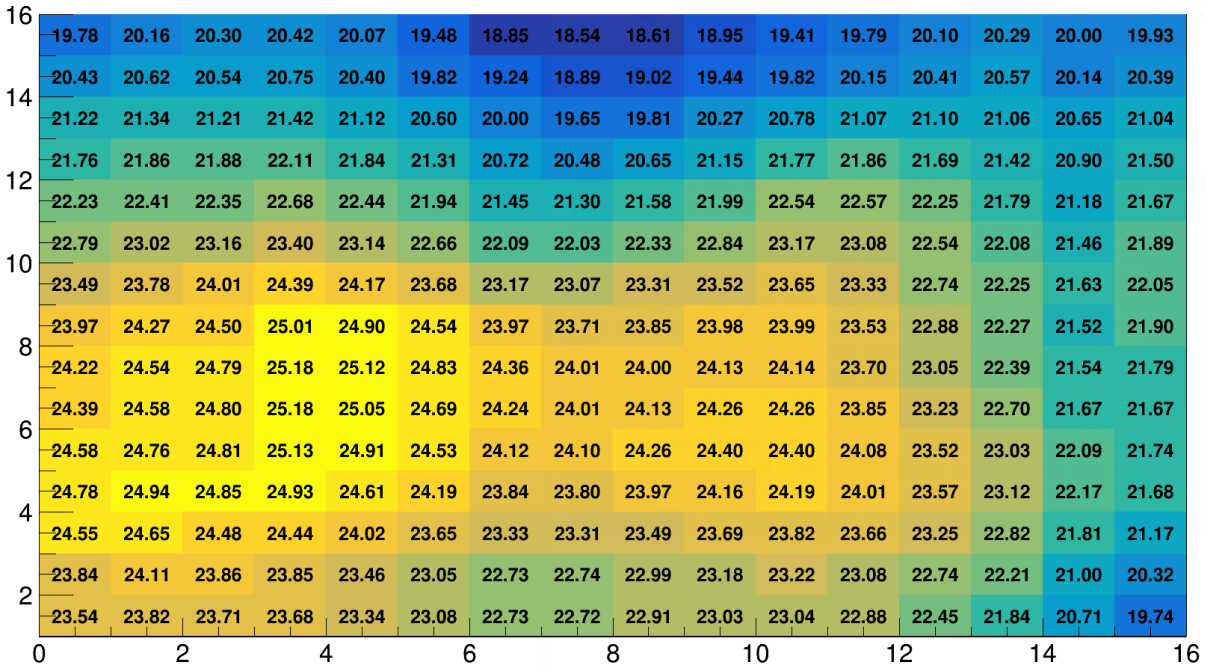


Figure 28: Pseudo-pixel thickness map for the 20-μm-thick sensor of the right telescope

8 Appendices

8.1 Peak localization algorithm

The "sliding window" peak localization algorithm consists of the following steps:

1. For each spectrum, the peak positions are determined by the peak the searching algorithm provided by ROOT (TSpectrum class):

$$\overline{X} = \{x_1,...,x_n\},$$

2. Determination of the initial position of the left boundary of the sliding window:

$$a_i = x_i - W_i \ ,$$

where a_i is the left boundary of the window, W_i is the width of the window predefined by the user.

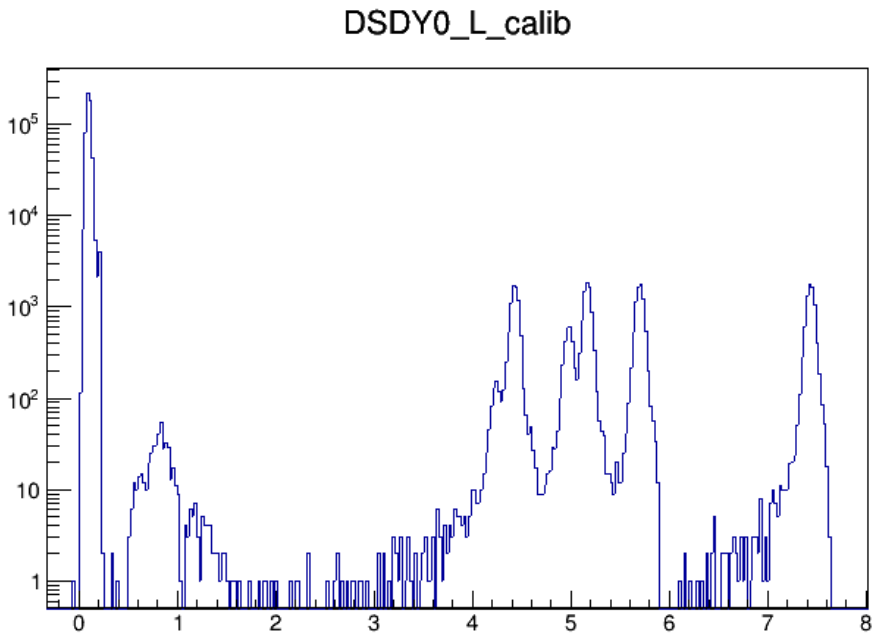
3. Determination of the final position of the left boundary of the sliding window (b_i):

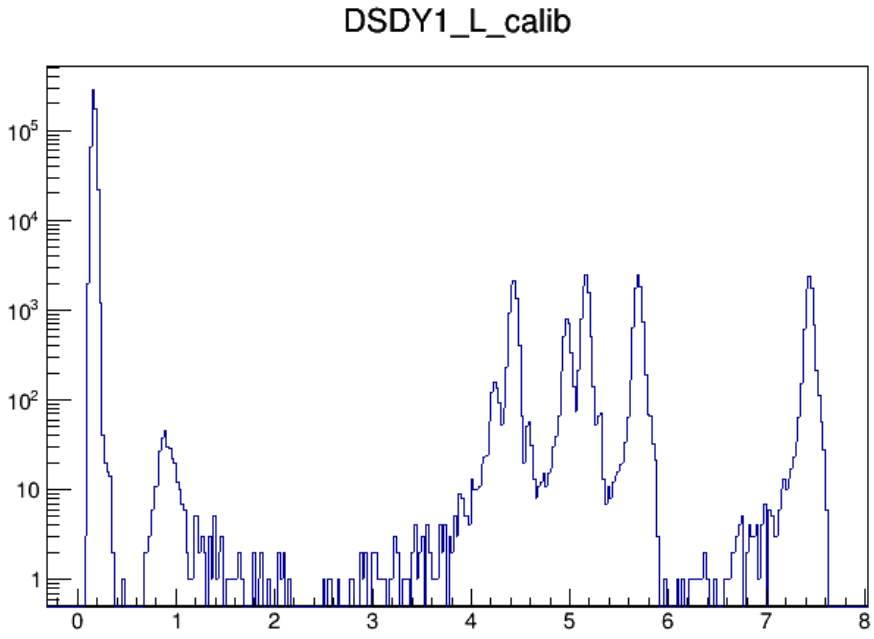
$$b_i = x_i,$$

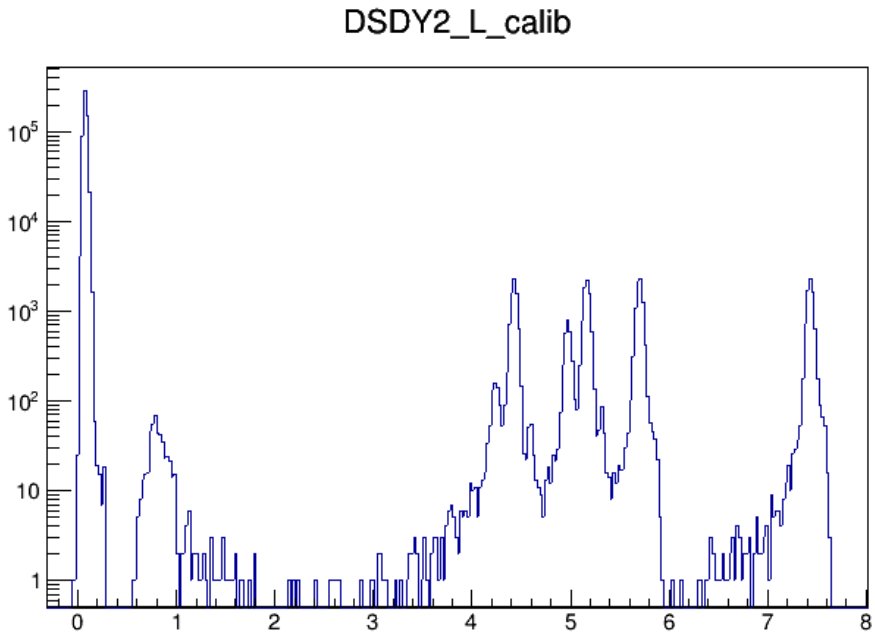
4. The window of the fixed width slides to the right with an increment of one bin. On each step, the integral in the window is counted. A position of the window with the maximum integral is used for calculation of the mean value (N_{mean}) which is considered as the peak position. RMS in the same window is also calculated and can be used in further analysis

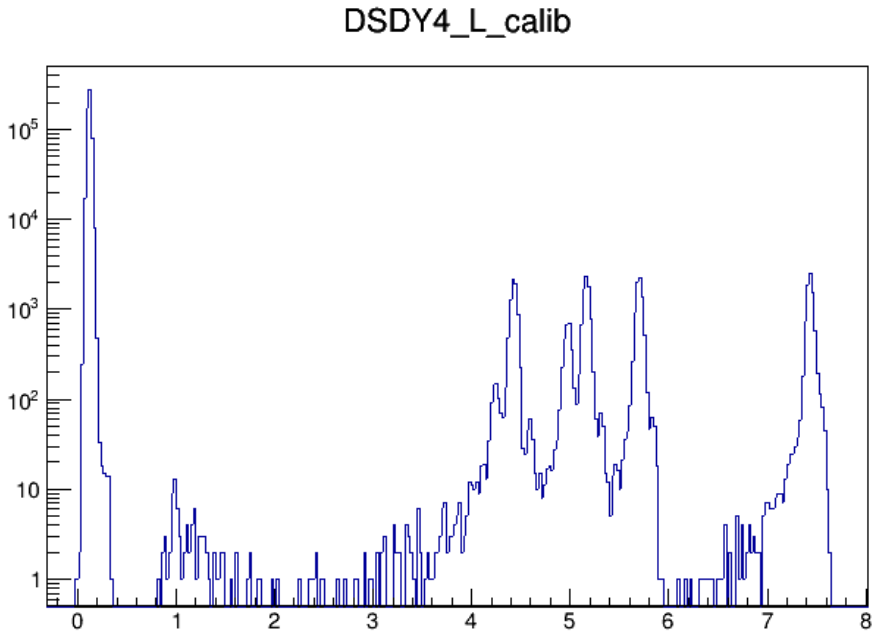
8.2 Calibrated spectra

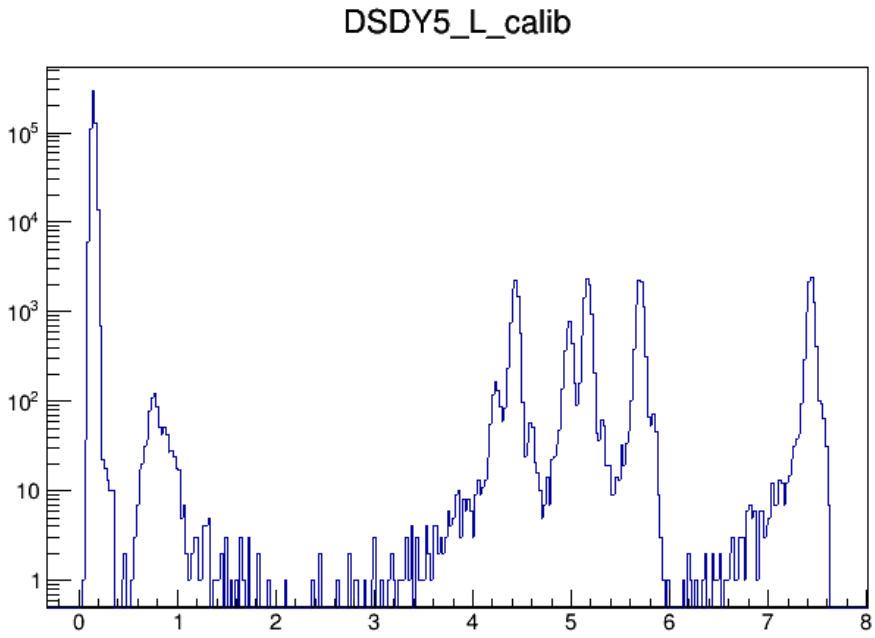
8.2.1 Left telescope

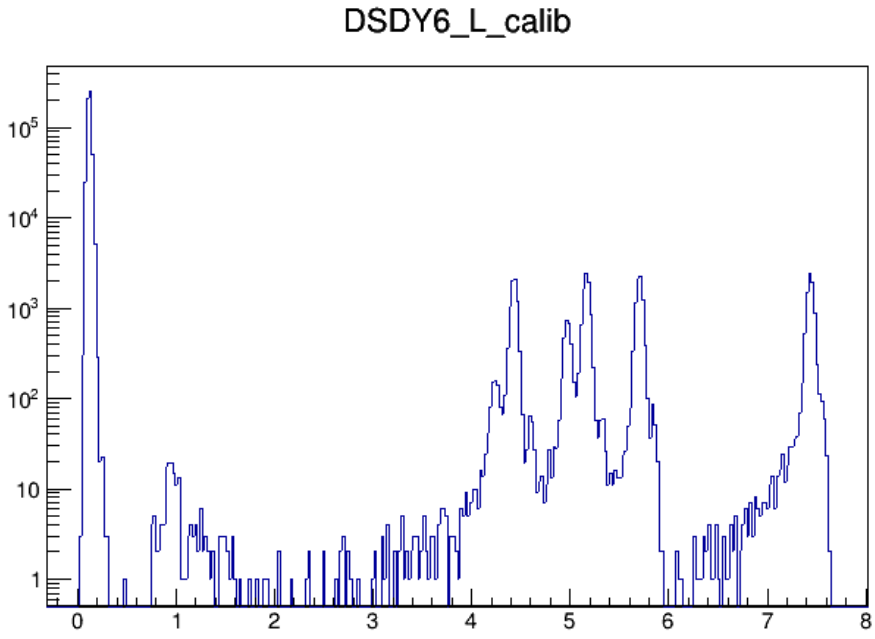


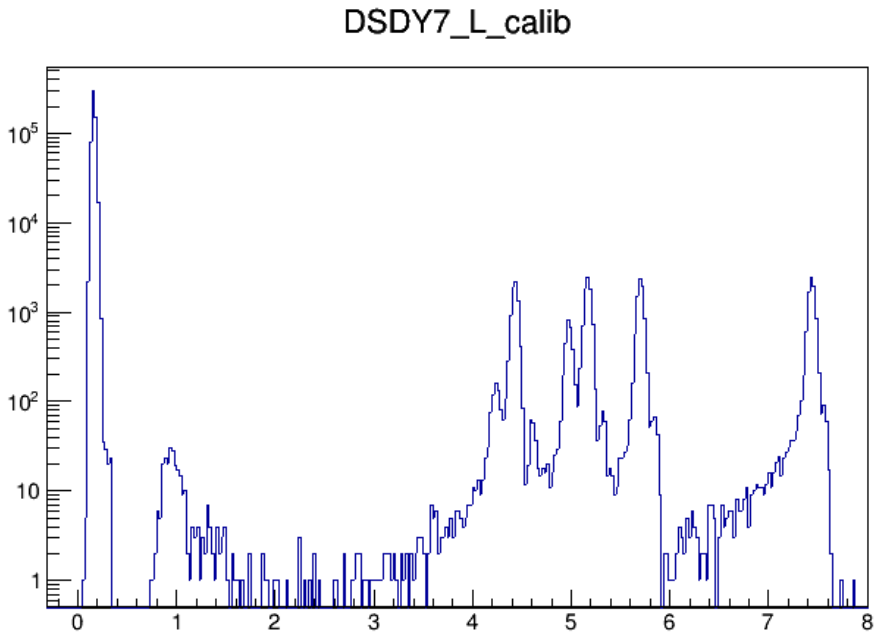

DSDY1_L_calib
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10^4
10^3
10^2
10
1
0
1
2
3
4
5
6
7
8

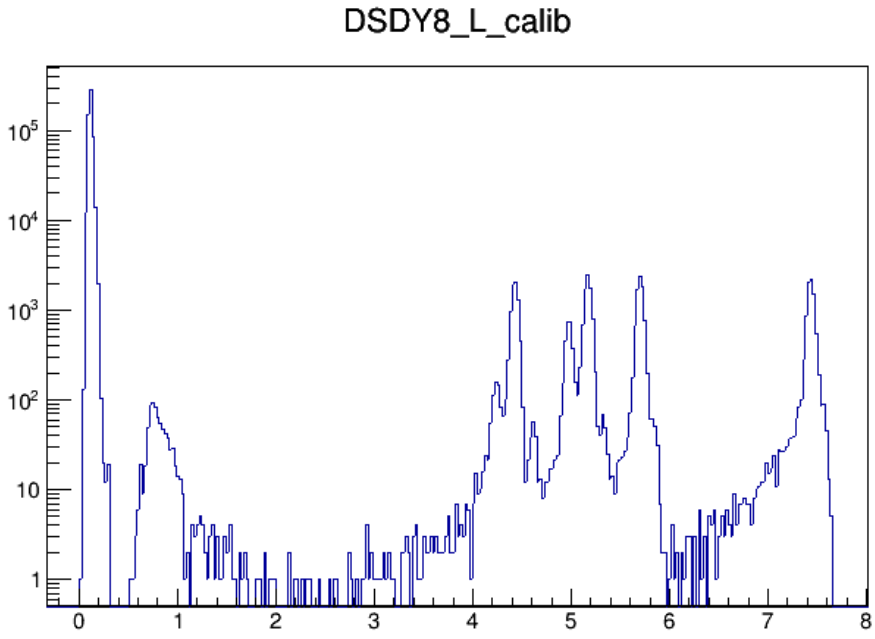

DSDY2_L_calib
10^5
10^4
10^3
10^2
10
1
0
1
2
3
4
5
6
7
8

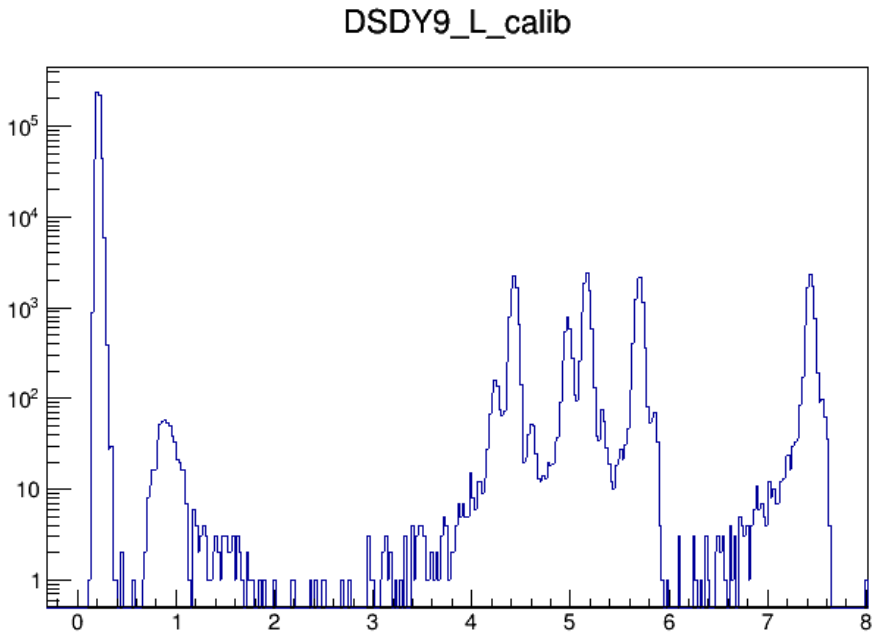

DSDY4_L_calib
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10^4
10^3
10^2
10
1
0
1
2
3
4
5
6
7
8

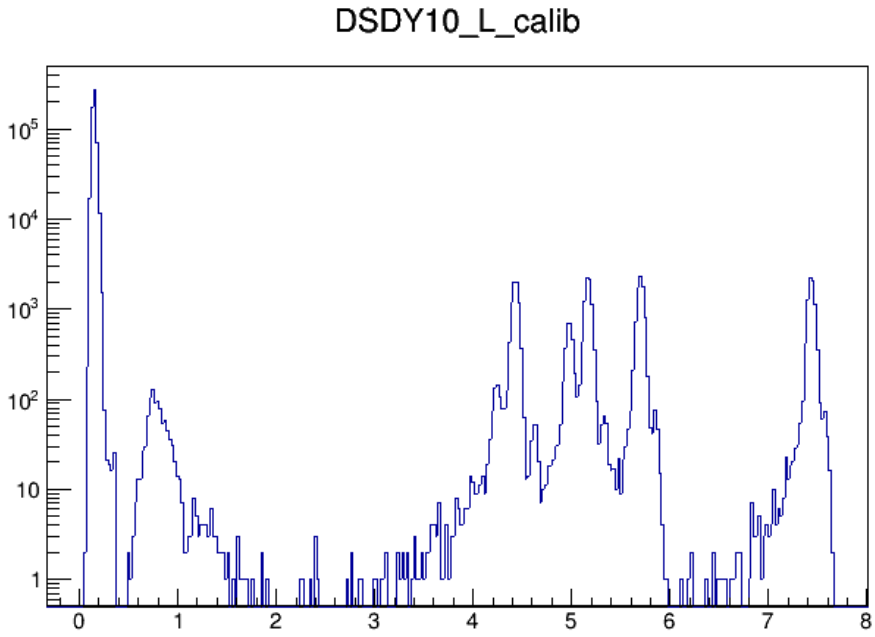

DSDY5_L_calib
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10^4
10^3
10^2
10
1
0
1
2
3
4
5
6
7
8

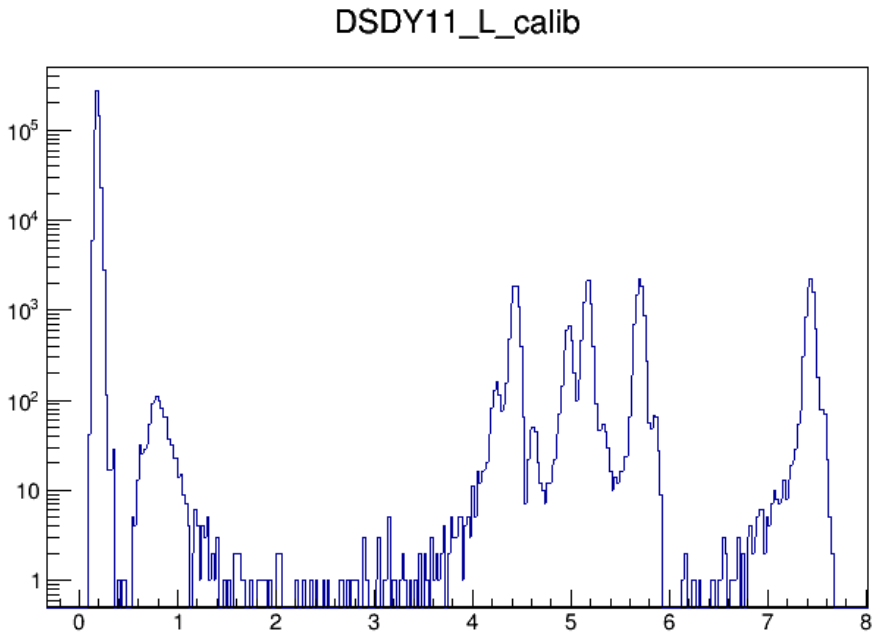

DSDY6_L_calib
10^5
10^4
10^3
10^2
10
1
0
1
2
3
4
5
6
7
8

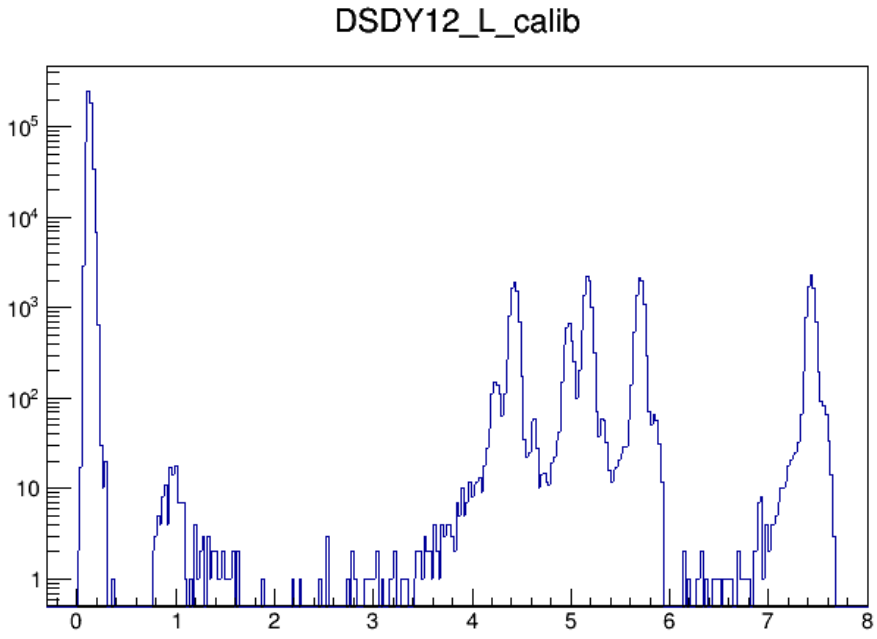

DSDY7_L_calib
10^5
10^4
10^3
10^2
10
1
0
1
2
3
4
5
6
7
8

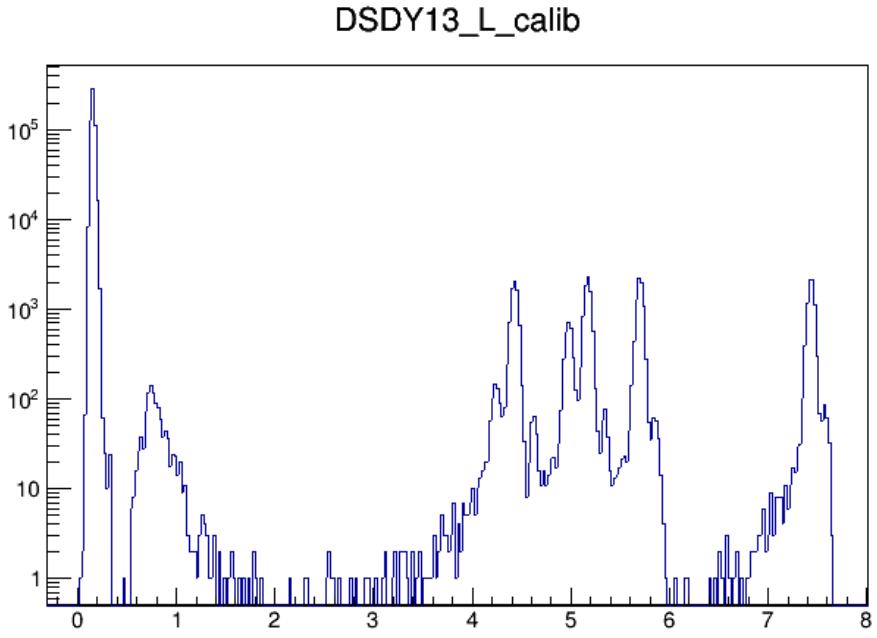

DSDY8_L_calib
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10^4
10^3
10^2
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1
0
1
2
3
4
5
6
7
8

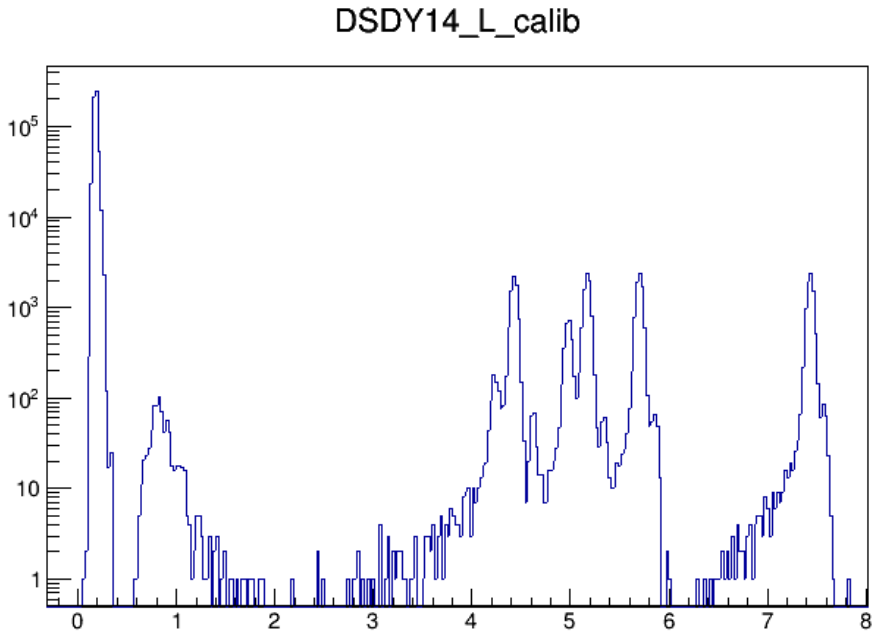

DSDY9_L_calib
10^5
10^4
10^3
10^2
10
1
0
1
2
3
4
5
6
7
8

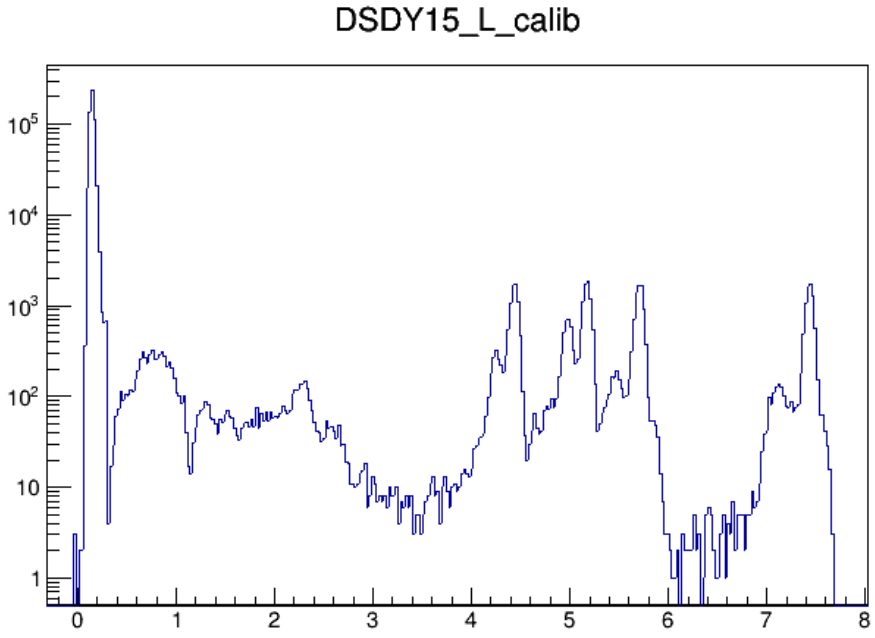

DSDY10_L_calib
10^5
10^4
10^3
10^2
10
1
0
1
2
3
4
5
6
7
8


DSDY11_L_calib
10^5
10^4
10^3
10^2
10
1
0
1
2
3
4
5
6
7
8

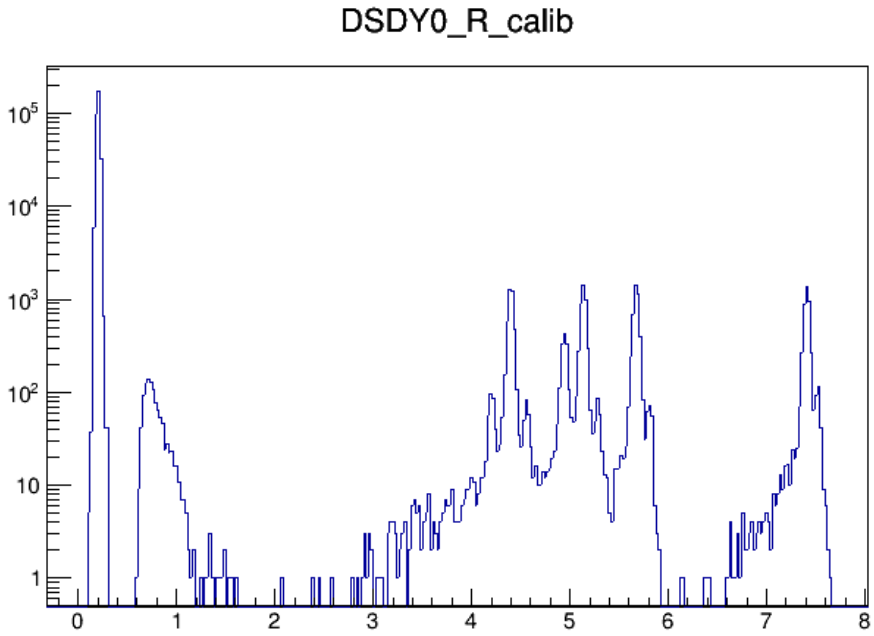

DSDY12_L_calib
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10^4
10^3
10^2
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1
0
1
2
3
4
5
6
7
8

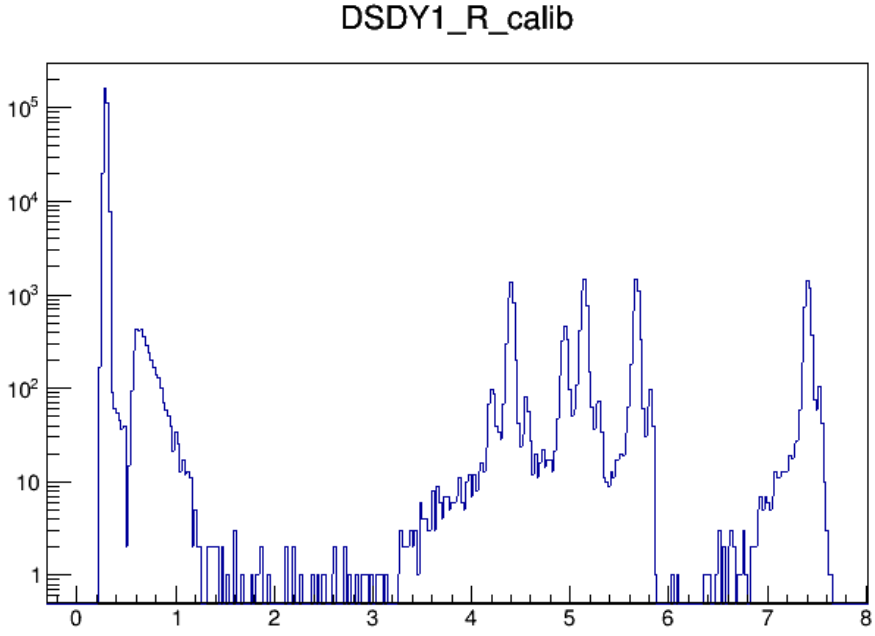

DSDY13_L_calib
10^5
10^4
10^3
10^2
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1
0
1
2
3
4
5
6
7
8

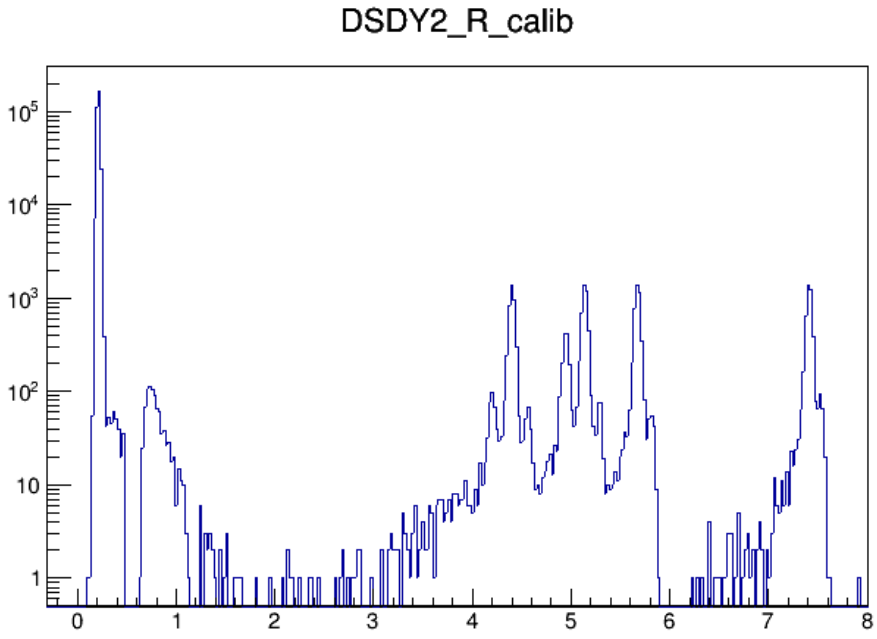

DSDY14_L_calib
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10^4
10^3
10^2
10
1
0
1
2
3
4
5
6
7
8

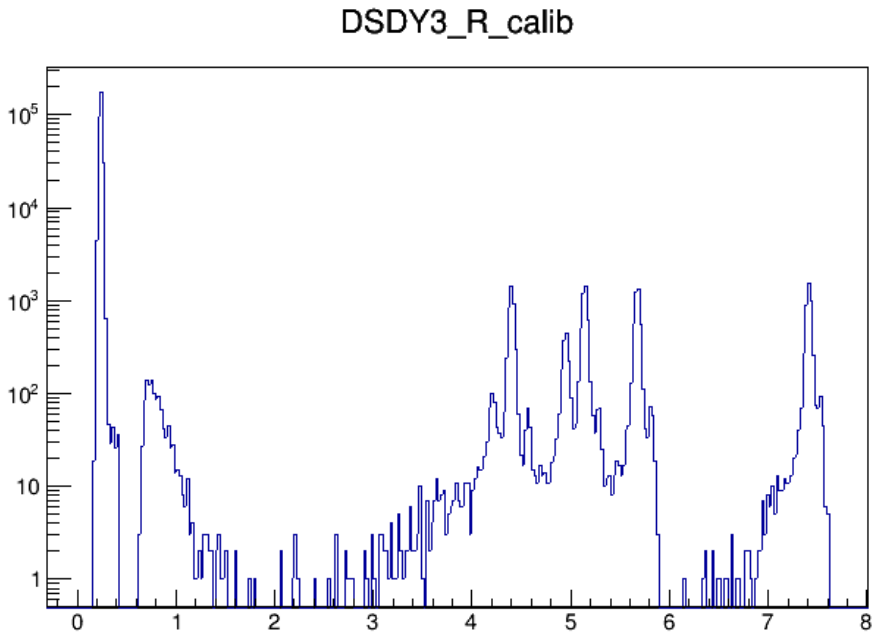

DSDY15_L_calib
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10^4
10^3
10^2
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1
0
1
2
3
4
5
6
7
8

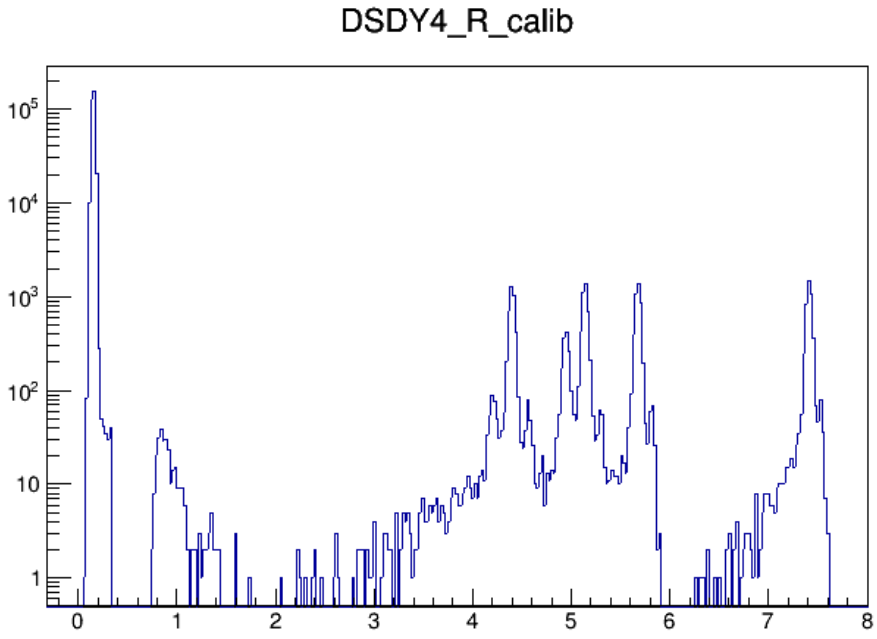
8.2.2 Right telescope

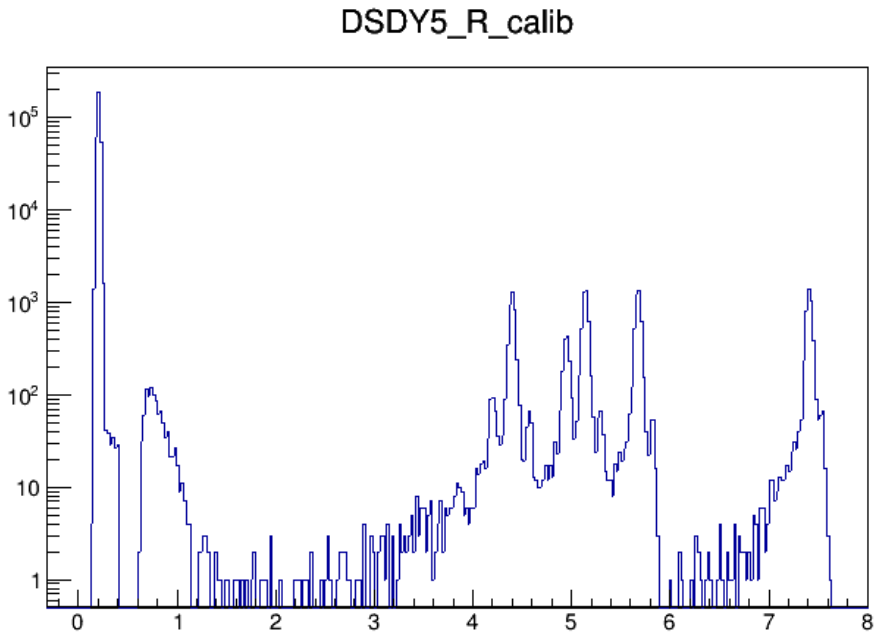


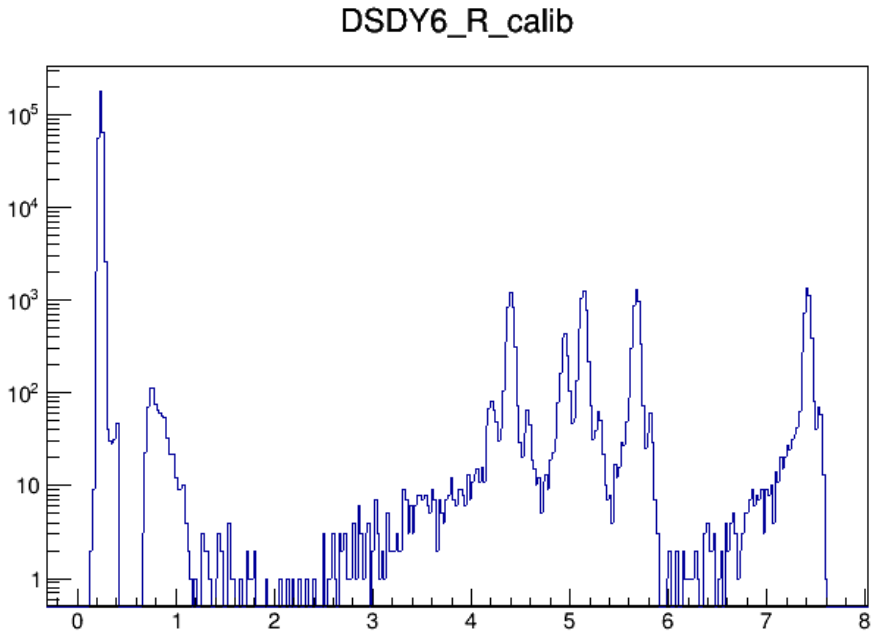


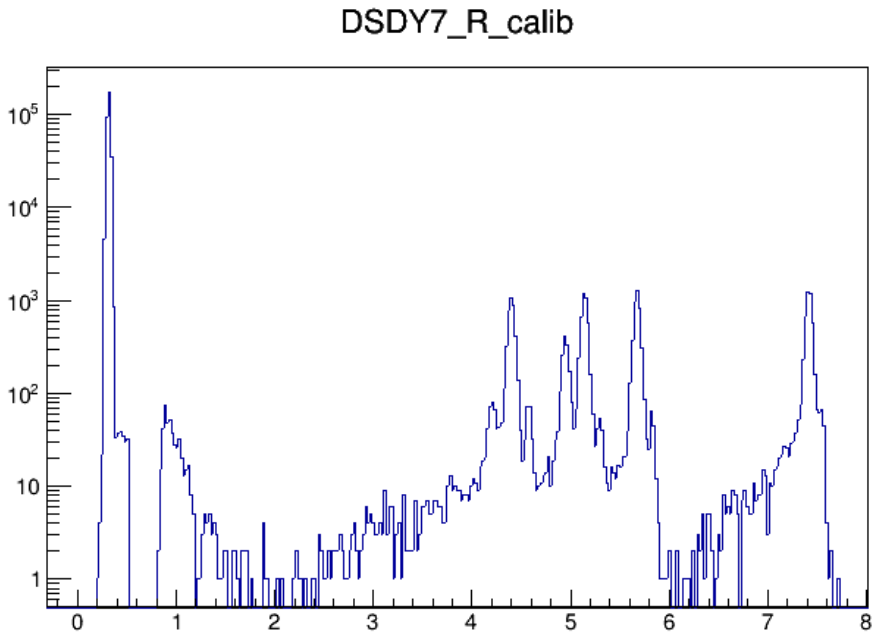

DSDY2_R_calib
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10^4
10^3
10^2
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1
2
3
4
5
6
7
8

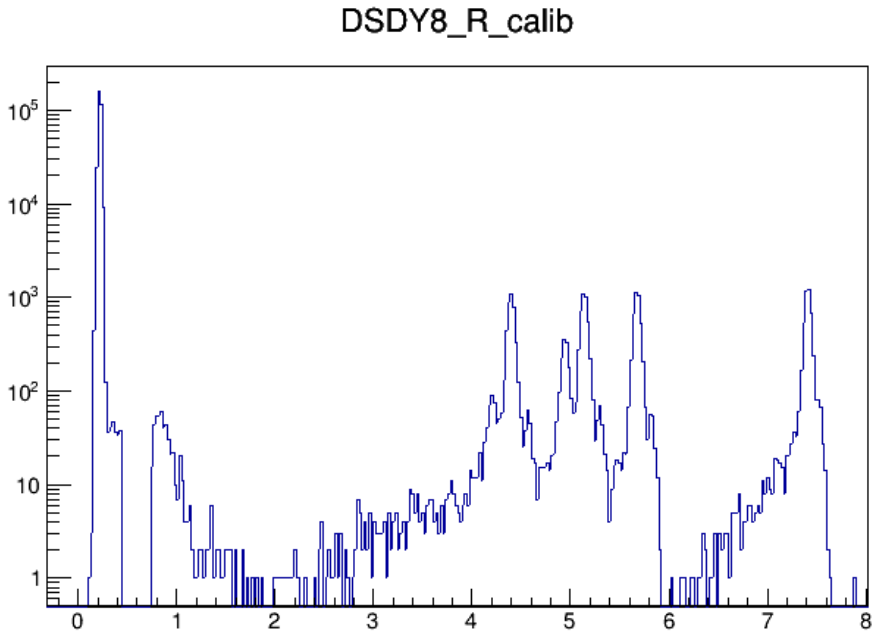

DSDY3_R_calib
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10^4
10^3
10^2
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2
3
4
5
6
7
8

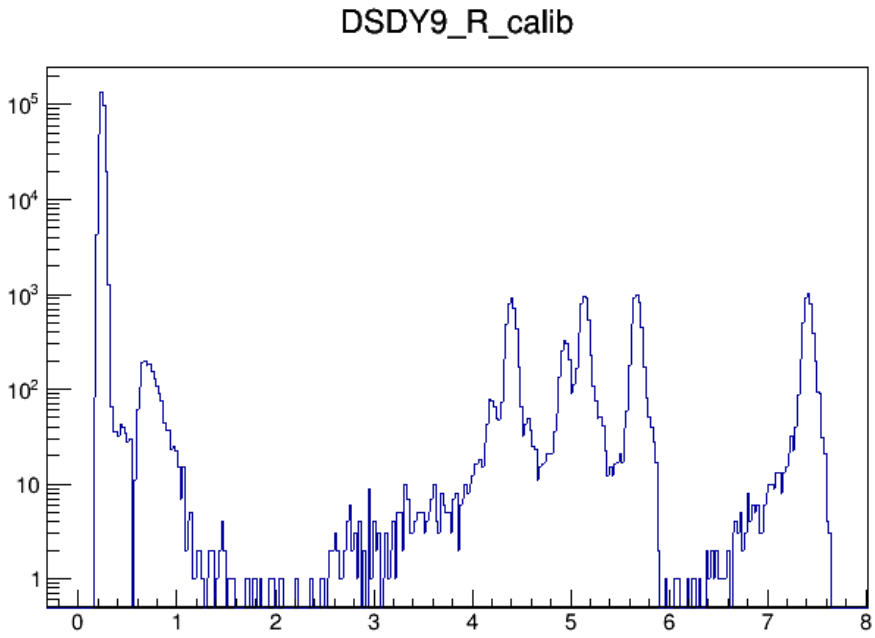

DSDY4_R_calib
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10^4
10^3
10^2
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1
0
1
2
3
4
5
6
7
8

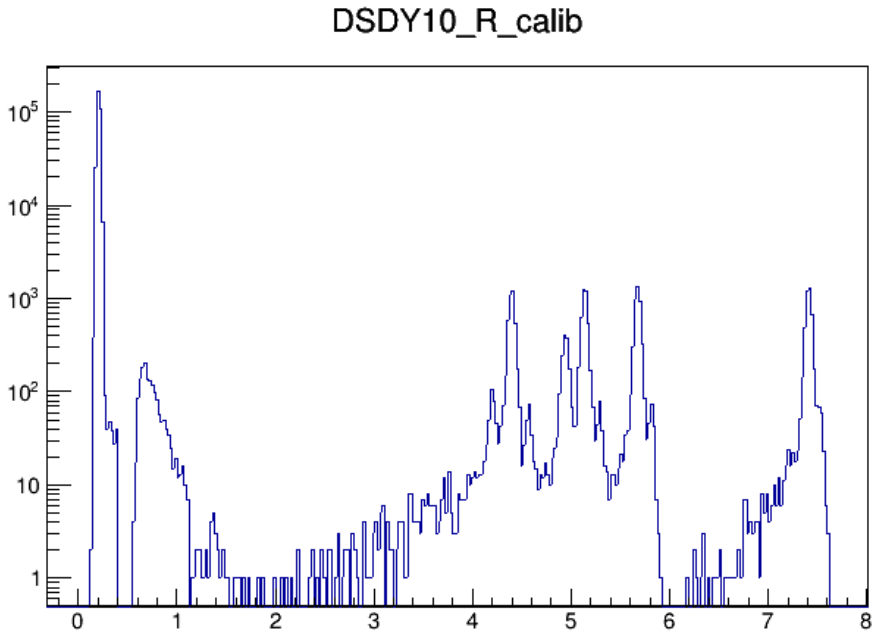

DSDY5_R_calib
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10^4
10^3
10^2
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2
3
4
5
6
7
8

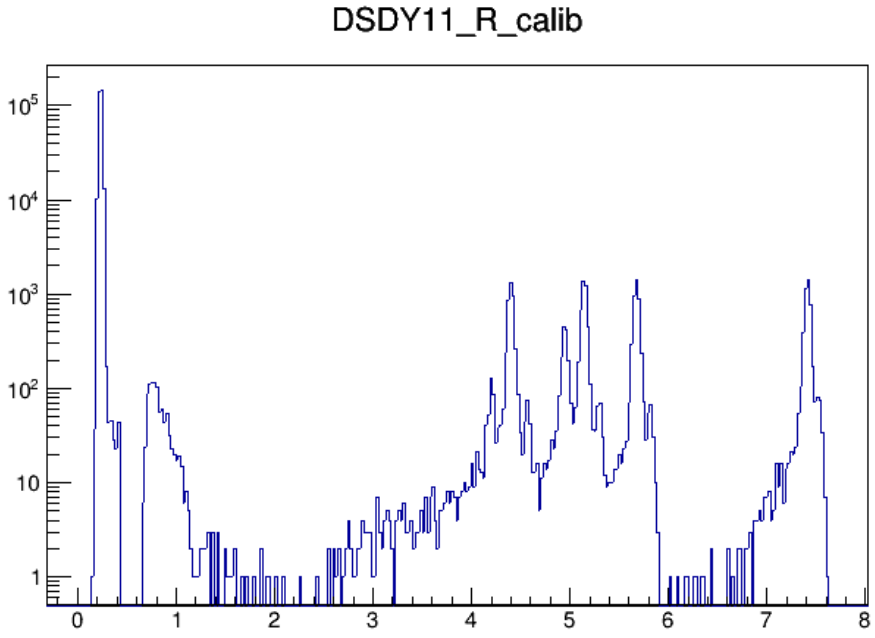

DSDY6_R_calib
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10^4
10^3
10^2
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1
2
3
4
5
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7
8

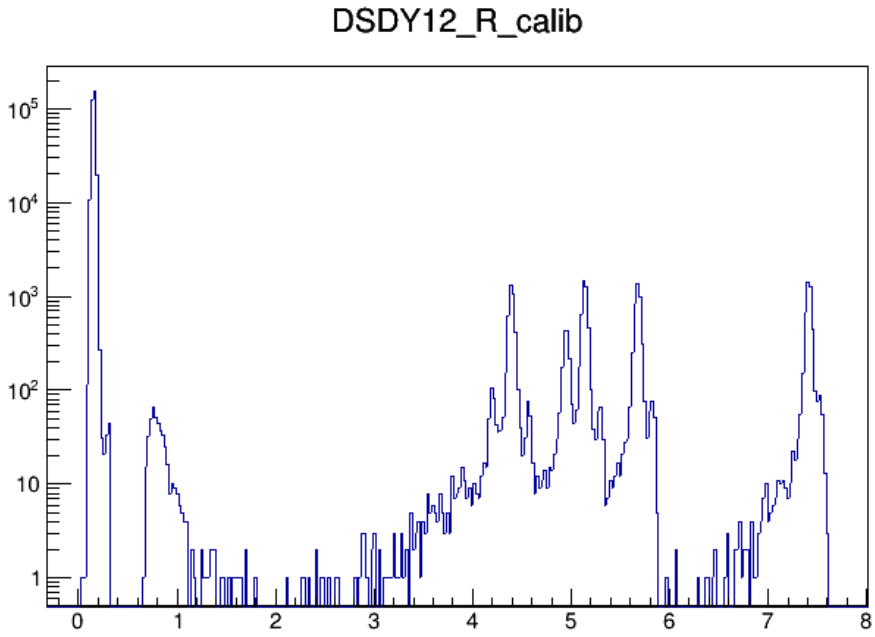

DSDY7_R_calib
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10^4
10^3
10^2
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0
1
2
3
4
5
6
7
8

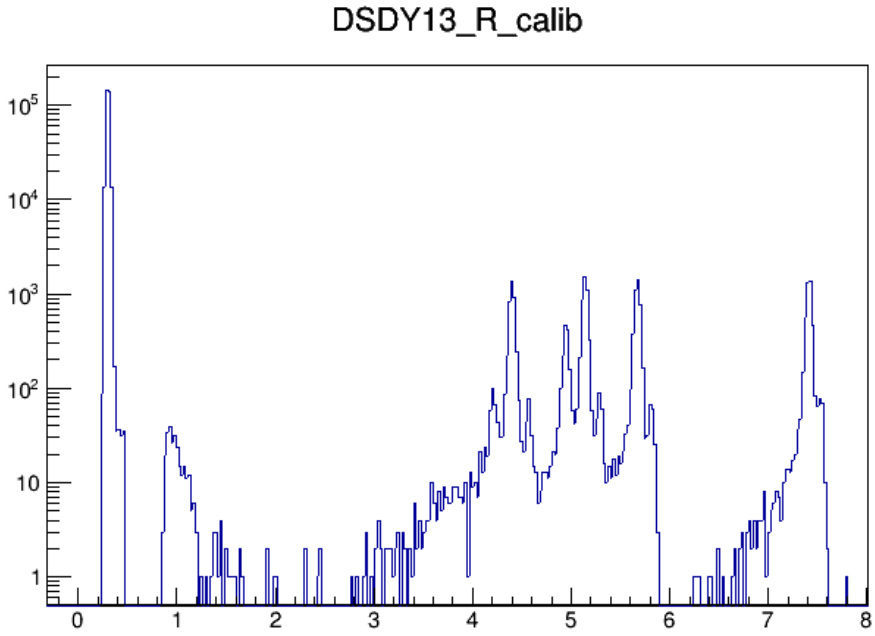

DSDY8_R_calib
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10^3
10^2
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1
2
3
4
5
6
7
8

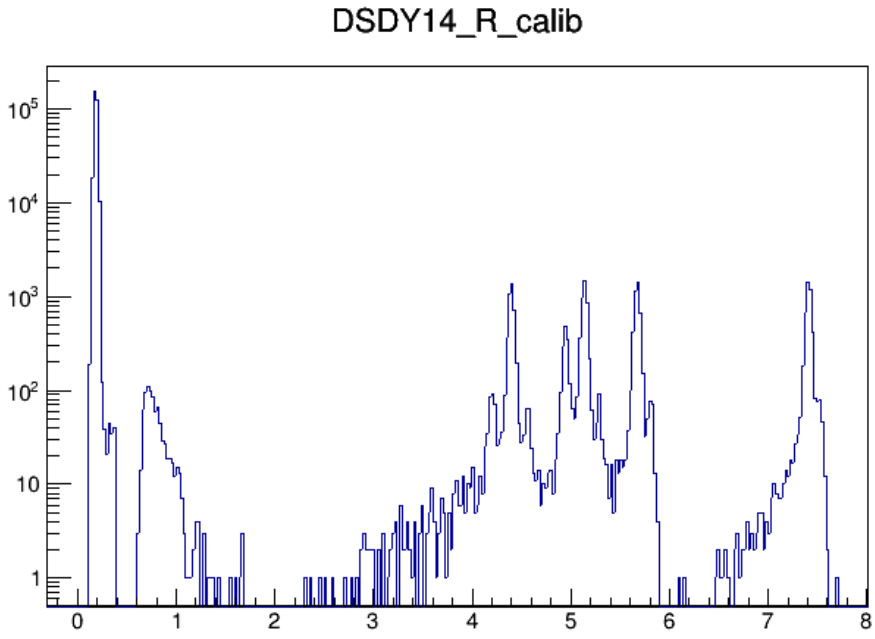

DSDY9_R_calib
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10^4
10^3
10^2
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1
2
3
4
5
6
7
8

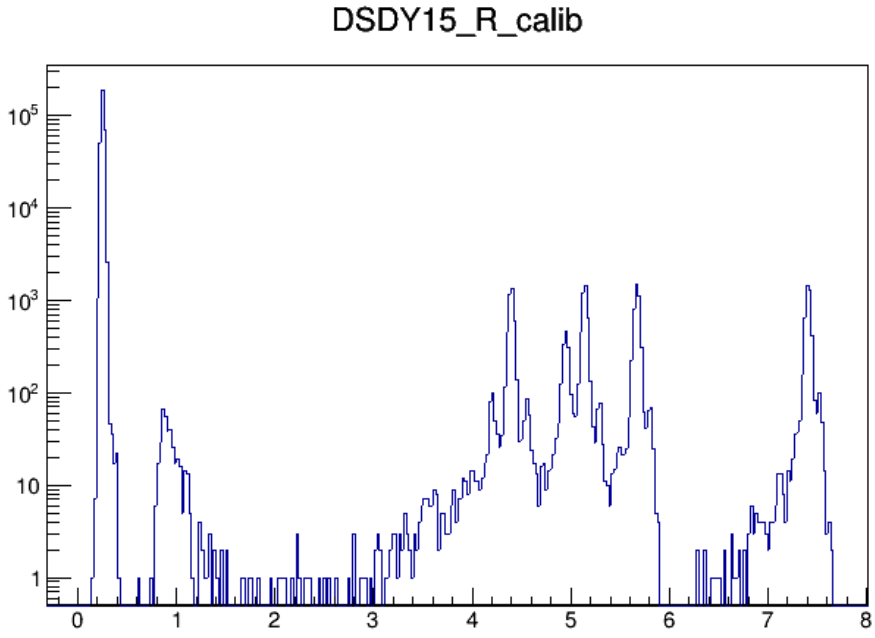

DSDY10_R_calib
10^5
10^4
10^3
10^2
10
1
0
1
2
3
4
5
6
7
8


DSDY11_R_calib
10^5
10^4
10^3
10^2
10
1
0
1
2
3
4
5
6
7
8


DSDY12_R_calib
10^5
10^4
10^3
10^2
10
1
0
1
2
3
4
5
6
7
8


DSDY13_R_calib
10^5
10^4
10^3
10^2
10
1
0
1
2
3
4
5
6
7
8


DSDY14_R_calib
10^5
10^4
10^3
10^2
10
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1
2
3
4
5
6
7
8


DSDY15_R_calib
10^5
10^4
10^3
10^2
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1
2
3
4
5
6
7
8